4

Linear relationships

Maths in context: Animation, cartoon, and game developers

The virtual world of animation has made computer gaming and cartoon videos into global multimillion dollar industries. Animation algorithms apply skills from algebra, coordinate geometry, linear relations, trigonometry, and more advanced mathematics.

When designing a cartoon character, the animator codes a subdivision process. Using the midpoint formula, a 2D polygon with straight sides is morphed into a curve. By repeatedly subdividing lengths and slicing off corners, a virtual curve illusion is created.

A similar subdivision process is coded to build 3D shapes. The initial polyhedral object has flat polygon faces that are then repeatedly subdivided and reduced. The finished shape has countless flat surfaces appearing on a computer screen as a smoothly curved virtual 3D object.

Linear algebra is used to control the position and movement when animating a virtual object. It can be transformed using linear equations that alter the coordinates of its vertices. Up or down movement is achieved by adding or subtracting, enlargement or shrinking uses multiplication or division, and rotation uses trigonometry.

The cartoon Bluey (pictured) is a successful pre-school TV animated cartoon developed in Brisbane and distributed internationally on Disney+.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Chapter contents

NSW Syllabus

In this chapter, a student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly (MAO-WM-01)
- determines the midpoint, gradient and length of an interval, and graphs linear relationships, with and without digital tools (MA5-LIN-C-01)
- graphs and interprets linear relationships using the gradient/slope-intercept form (MA5-LIN-C-02)
- describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems (MA5-LIN-P-01).

© 2022 NSW Education Standards Authority

Online resources

A host of additional online resources are included as part of your Interactive Textbook, including HOTmaths content, video demonstrations of all worked examples, auto-marked quizzes and much more.

4A Introducing linear relationships CONSOLIDATING

Learning intentions

- To review the features of the Cartesian plane and plot points using coordinates
- To know that a linear relationship gives a set of points that form a straight line
- To be able to complete a table of values and plot points to form a linear graph
- To be able to arrange the equation of a line in various ways, including general form
- To be able to decide if a point is on a line by substituting into the equation of the line
- To know which points are represented by the x-intercept and y-intercept
- To be able to identify the x-intercept and y-intercept from a table or graph

Past, present and future learning

- This section consolidates Stage 4 concepts which are used in Stages 5 and 6
- Some of these questions are more challenging than those in the Year 8 book
- These concepts will be revised and extended in Chapter 1 of our Year 10 book
- Expertise with these concepts is assumed knowledge for Stage 6 Advanced

If two variables are related in some way, we can use mathematical rules to precisely describe this relationship. The most simple kind of mathematical relationship is one that can be illustrated with a straight line graph. These are called linear relationships. The volume of petrol in your car at a service bowser, for example, might initially be 10 L then be increasing by 1.2 L per second after that. This is an example of a linear relationship between *volume* and *time* because the volume is increasing at a constant rate of 1.2 L/s.

A car's annual running costs can be \$4000 p.a., including fuel, repairs, registration and insurance. If a car cost \$17 000, the total cost, C, for n years could be expressed as the linear relationship: $C = 4000n + 17000$.

Lesson starter: Is it linear?

Here are three rules linking x and y.

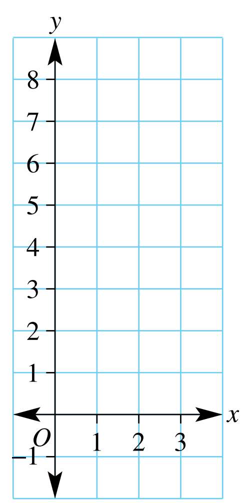
1 $y_1 = \frac{2}{x} + 1$

2 $y_2 = x^2 - 1$

3 $y_3 = 3x - 4$

First complete this simple table and graph.

x	1	2	3
y_1			
y_2			
y_3			

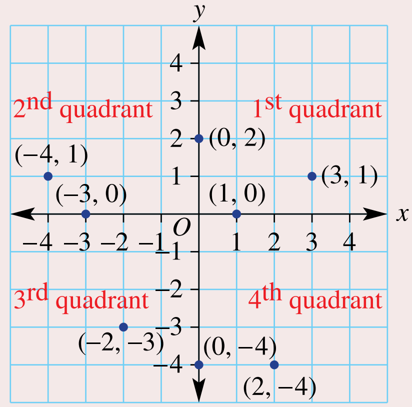


- Which of the three rules do you think is linear?
- How do the table and graph help you decide it's linear?

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

- **Coordinate geometry** provides a link between geometry and algebra.
- The **Cartesian plane** (or number plane) consists of two axes which divide the number plane into four **quadrants**.
  - The horizontal x-axis and vertical y-axis intersect at right angles at the **origin** $O(0, 0)$.
  - A point is precisely positioned on a Cartesian plane using the **coordinate pair** (x, y) where x describes the horizontal position and y describes the vertical position of the point from the origin.

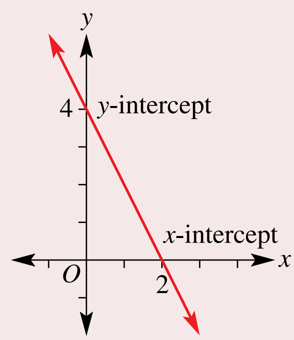


- A **linear relationship** is a set of ordered pairs (x, y) that when graphed give a straight line.
- Linear relationships have rules that may be of the form:
  - $y = mx + c$. For example, $y = 2x + 1$.
  - $ax + by = d$ or $ax + by + c = 0$. For example, $2x - 3y = 4$ or $2x - 3y - 4 = 0$.
- The x-intercept is the point where the graph meets the x-axis.
- The y-intercept is the point where the graph meets the y-axis.

x-intercept

x	−2	−1	0	1	2	3
y	8	6	4	2	0	−2

y-intercept



BUILDING UNDERSTANDING

1 Refer to the Cartesian plane shown.

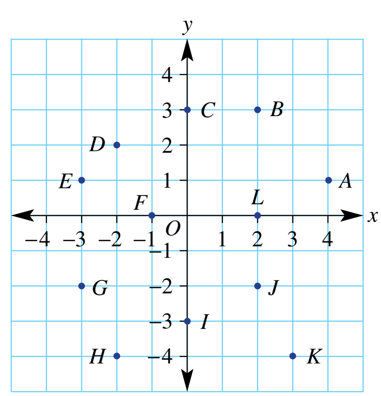
a Give the coordinates of all the points $A - L$.

b State which points lie on:

i the x-axis **ii** the y-axis.

c State which points lie inside the:

i 2nd quadrant **ii** 4th quadrant.



2 For the rule $y = 3x - 4$, find the value of y for these x-values.

a $x = 2$ **b** $x = -1$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

3 For the rule $y = -2x + 1$, find the value of y for these x-values.

a $x = 0$

b $x = -10$

4 Give the coordinates of three pairs of points on the line $y = x + 3$.

Example 1 Plotting points to graph straight lines

Using $-3 \leqslant x \leqslant 3$, construct a table of values and plot a graph for these linear relationships.

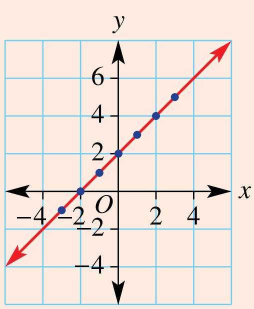
a $y = x + 2$

b $y = -2x + 2$

SOLUTION

a

x	−3	−2	−1	0	1	2	3
y	−1	0	1	2	3	4	5



EXPLANATION

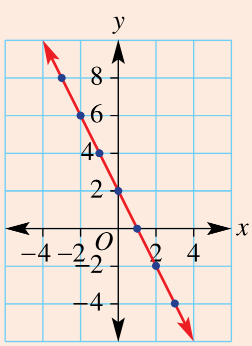
Use $-3 \leqslant x \leqslant 3$ as instructed and substitute each value of x into the rule $y = x + 2$.

The coordinates of the points are read from the table, i.e. $(-3, -1)$, $(-2, 0)$, etc.

Plot each point and join to form a straight line. Extend the line to show it continues in either direction.

b

x	−3	−2	−1	0	1	2	3
y	8	6	4	2	0	−2	−4



Use $-3 \leqslant x \leqslant 3$ as instructed and substitute each value of x into the rule $y = -2x + 2$.

For example:

$$\begin{aligned} x = -3,\ y &= -2 \times (-3) + 2 \\ &= 6 + 2 \\ &= 8 \end{aligned}$$

Plot each point and join to form a straight line. Extend the line beyond the plotted points.

Now you try

Using $-3 \leqslant x \leqslant 3$, construct a table of values and plot a graph for these linear relationships.

a $y = x - 1$

b $y = -3x + 1$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

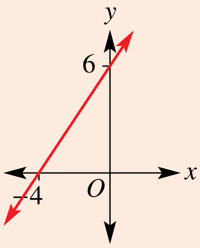
Example 2 Reading off the x-intercept and y-intercept

Write down the coordinates of the x-intercept and y-intercept from this table and graph.

a

x	−2	−1	0	1	2	3
y	6	4	2	0	−2	−4

b



SOLUTION

a The x-intercept is at $(1, 0)$.

The y-intercept is at $(0, 2)$.

b The x-intercept is at $(-4, 0)$.

The y-intercept is at $(0, 6)$.

EXPLANATION

The x-intercept is at the point where $y = 0$ (on the x-axis).

The y-intercept is at the point where $x = 0$ (on the y-axis).

The x-intercept is at the point where $y = 0$ (on the x-axis).

The y-intercept is at the point where $x = 0$ (on the y-axis).

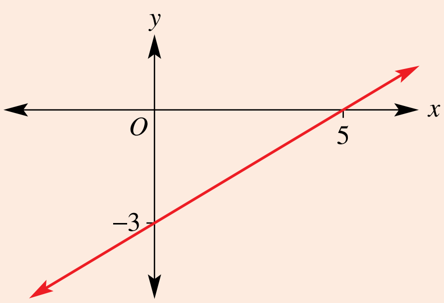
Now you try

Write down the coordinates of the x-intercept and y-intercept from this table and graph.

a

x	−3	−2	−1	0	1	2
y	4	3	2	1	0	−1

b



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 3 Deciding if a point is on a line

Decide if the point $(-2, 4)$ is on the line with the given rules.

a $y = 2x + 10$ **b** $y = -x + 2$

SOLUTION

a $y = 2x + 10$

Substitute $x = -2$

$y = 2(-2) + 10$

$= 6$

$\therefore$ the point $(-2, 4)$ is not on the line.

b $y = -x + 2$

Substitute $x = -2$

$y = -(-2) + 2$

$= 4$

$\therefore$ the point $(-2, 4)$ is on the line.

EXPLANATION

a Find the value of y by substituting $x = -2$ into the rule for y.

The y-value is not 4, so $(-2, 4)$ is not on the line.

b By substituting $x = -2$ into the rule for the line, y is 4.

So $(-2, 4)$ is on the line.

Now you try

Decide if the point $(-1, 6)$ is on the line with the given rules.

a $y = -2x + 4$ **b** $y = 3x + 5$

Exercise 4A

FLUENCY 1–3(½) 1–4(½) 1–4(½)

Example 1

1 Using $-3 \leqslant x \leqslant 3$, construct a table of values and plot a graph for these linear relations.

a $y = x - 1$ **b** $y = 2x - 3$ **c** $y = -x + 4$ **d** $y = -3x$

Example 2

2 Write down the coordinates of the x- and y-intercepts for the following tables and graphs.

a

x	−3	−2	−1	0	1	2	3
y	4	3	2	1	0	−1	−2

b

x	−3	−2	−1	0	1	2	3
y	−1	0	1	2	3	4	5

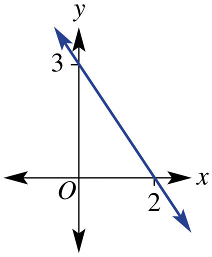
c

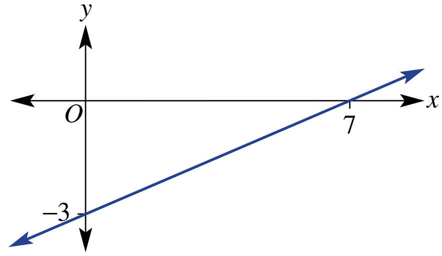
x	−1	0	1	2	3	4
y	10	8	6	4	2	0

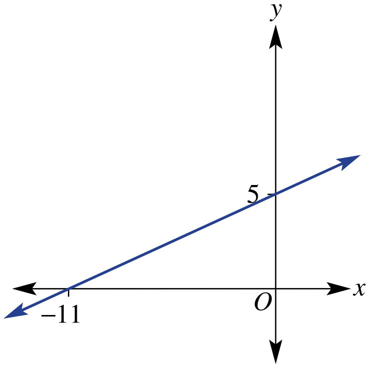
d

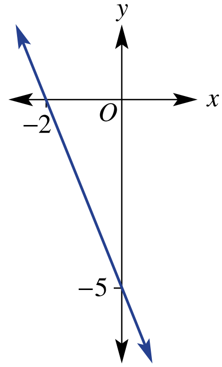
x	−5	−4	−3	−2	−1	0
y	0	2	4	6	8	10

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

e


f


g


h


Example 3

3 Decide if the point (1, 2) is on the line with the given rule.

a $y = x + 1$ **b** $y = 2x - 1$ **c** $y = -x + 3$

d $y = -2x + 4$ **e** $y = -x + 5$ **f** $y = -\frac{1}{2}x + \frac{1}{2}$

4 Decide if the point (−3, 4) is on the line with the given rule.

a $y = 2x + 8$ **b** $y = x + 7$ **c** $y = -x - 1$

d $y = \frac{1}{3}x + 6$ **e** $y = -\frac{1}{3}x + 3$ **f** $y = \frac{5}{6}x + \frac{11}{2}$

PROBLEM–SOLVING | 5 | 5, 6 | 5(½), 6

5 Find a rule in the form $y = mx + c$ (e.g. $y = 2x - 1$) that matches these tables of values.

a

x	0	1	2	3	4
y	2	3	4	5	6

b

x	−1	0	1	2	3
y	−2	0	2	4	6

c

x	−2	−1	0	1	2	3
y	−3	−1	1	3	5	7

d

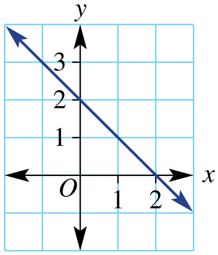
x	−3	−2	−1	0	1	2
y	5	4	3	2	1	0

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

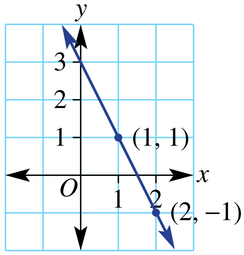
6 Match rules **a**, **b**, **c** and **d** to the graphs **A**, **B**, **C** and **D**.

a $y = 2x + 1$ **b** $y = -x - 1$ **c** $y = -2x + 3$ **d** $x + y = 2$

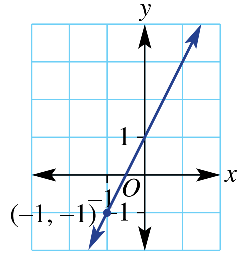
A



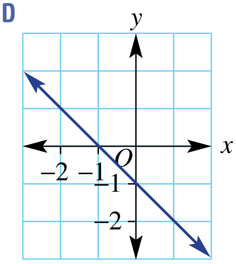
B



C



D



REASONING 7 | 7, 8 | 8, 9

7 Decide if the following rules are equivalent.

a $y = 1 - x$ and $y = -x + 1$

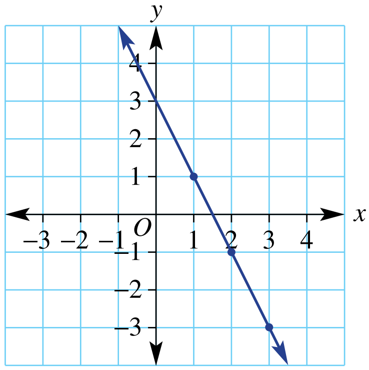
b $y = 1 - 3x$ and $y = 3x - 1$

c $y = -2x + 1$ and $y = -1 - 2x$

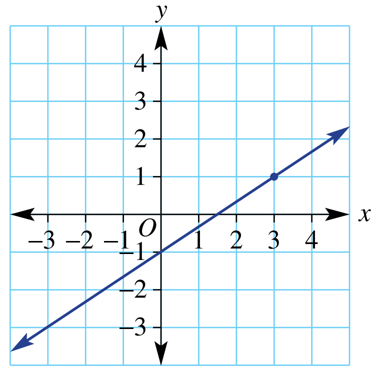
d $y = -3x + 1$ and $y = 1 - 3x$

8 Give reasons why the x-intercept on these graphs is at $\left(\frac{3}{2}, 0\right)$.

a



b



9 Decide if the following equations are true or false.

a $\frac{2x + 4}{2} = x + 4$

b $\frac{3x - 6}{3} = x - 2$

c $\frac{1}{2}(x - 1) = \frac{1}{2}x - \frac{1}{2}$

d $\frac{2}{3}(x - 6) = \frac{2}{3}x - 12$

ENRICHMENT: Tough rule finding – | – | 10

10 Find the linear rule linking x and y in these tables.

a

x	−1	0	1	2	3
y	5	7	9	11	13

b

x	−2	−1	0	1	2
y	22	21	20	19	18

c

x	0	2	4	6	8
y	−10	−16	−22	−28	−34

d

x	−5	−4	−3	−2	−1
y	29	24	19	14	9

e

x	1	3	5	7	9
y	1	2	3	4	5

f

x	−14	−13	−12	−11	−10
y	$5\frac{1}{2}$	5	$4\frac{1}{2}$	4	$3\frac{1}{2}$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4B Graphing straight lines using intercepts

Learning intentions
- To understand that only two points are required to graph a straight line
- To be able to graph a line by finding the *x*-intercept and *y*-intercept

Past, present and future learning
- This section addresses the Stage 5 Path Topic called *Linear relationships C*
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

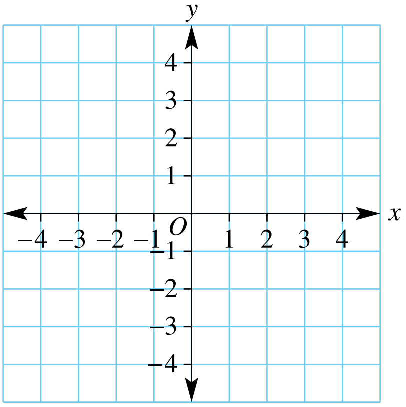
When linear rules are graphed, all the points lie in a straight line. It is therefore possible to graph a straight line using only two points. Two critical points that help draw these graphs are the x-intercept and y-intercept introduced in the previous section.

Businesses claim a tax deduction for the annual decrease in the value, V, of equipment over n years. A straight line segment joining the intercepts $(0, V)$ and $(n, 0)$ shows that the rate of decrease per year is $\frac{V}{n}$.

Lesson starter: Two key points

Consider the relation $y = \frac{1}{2}x + 1$ and complete this table and graph.

x	−4	−3	−2	−1	0	1	2
y							



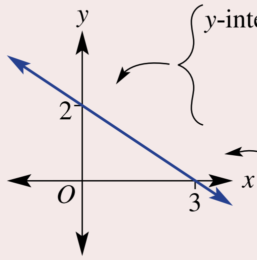
- What are the coordinates of the point where the line crosses the y-axis? That is, state the coordinates of the y-intercept.
- What are the coordinates of the point where the line crosses the x-axis? That is, state the coordinates of the x-intercept.
- Discuss how you might find the coordinates of the x- and y-intercepts without drawing a table and plotting points. Explain your method.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

- Two points are required to sketch a straight line graph. Often these points are the axes intercepts.
- The *y*-**intercept** is the point where the line intersects the *y*-axis (where $x = 0$).
  - Substitute $x = 0$ to find the *y*-intercept.
- The *x*-**intercept** is the point where the line intersects the *x*-axis (where $y = 0$).
  - Substitute $y = 0$ to find the *x*-intercept.

Example: $2x + 3y = 6$



y-intercept ($x = 0$)
$$2(0) + 3y = 6$$
$$3y = 6$$
$$y = 2$$

x-intercept ($y = 0$)
$$2x + 3(0) = 6$$
$$2x = 6$$
$$x = 3$$

BUILDING UNDERSTANDING

1 Mark the following *x*- and *y*-intercepts on a set of axes and join in a straight line.
- **a** *x*-intercept: $(-2, 0)$, *y*-intercept: $(0, 3)$
- **b** *x*-intercept: $(4, 0)$, *y*-intercept: $(0, 6)$

2 **a** Find the value of *y* in these equations.
- **i** $2y = 6$ **ii** $y = 3 \times 0 + 4$ **iii** $-2y = 12$

b Find the value of *x* in these equations.
- **i** $-4x = -40$ **ii** $0 = 2x - 2$ **iii** $\frac{1}{2}x = 3$

3 For these equations find the coordinates of the *y*-intercept by letting $x = 0$.
- **a** $x + y = 4$ **b** $x - y = 5$ **c** $2x + 3y = 9$

4 For these equations find the coordinates of the *x*-intercept by letting $y = 0$.
- **a** $2x - y = -4$ **b** $4x - 3y = 12$ **c** $y = 3x - 6$

Example 4 Sketching with intercepts

Sketch the graph of the following, showing the *x*- and *y*-intercepts.
- **a** $2x + 3y = 6$
- **b** $y = 2x - 6$

SOLUTION

a $2x + 3y = 6$

y-intercept (let $x = 0$):
$$2(0) + 3y = 6$$
$$3y = 6$$
$$y = 2$$

$\therefore$ the *y*-intercept is at $(0, 2)$.

EXPLANATION

Only two points are required to generate a straight line. For the *y*-intercept, substitute $x = 0$ into the rule and solve for *y* by dividing each side by 3.

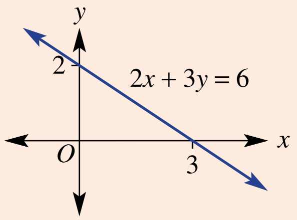
State the *y*-intercept.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

x-intercept (let $y = 0$):
$2x + 3(0) = 6$
$2x = 6$
$x = 3$
$\therefore$ the x-intercept is at $(3, 0)$.

Similarly to find the x-intercept, substitute $y = 0$ into the rule and solve for x.

State the x-intercept.



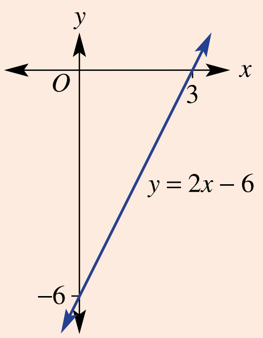
Mark and label the intercepts on the axes and sketch the graph by joining the two intercepts.

b $y = 2x - 6$
y-intercept (let $x = 0$):
$y = 2(0) - 6$
$y = -6$
$\therefore$ the y-intercept is at $(0, -6)$.

Substitute $x = 0$ for the y-intercept.
Simplify to find the y-coordinate.

x-intercept (let $y = 0$):
$0 = 2x - 6$
$6 = 2x$
$x = 3$
$\therefore$ the x-intercept is at $(3, 0)$.

Substitute $y = 0$ for the x-intercept. Solve the remaining equation for x by adding 6 to both sides and then divide both sides by 2.



Mark in the two intercepts and join to sketch the graph.

Now you try

Sketch the graph of the following, showing the x- and y-intercepts.

a $3x + 5y = 15$ **b** $y = 3x - 6$

Exercise 4B

FLUENCY	1–2(½)	1–3(½)	1–3(⅓)

Example 4a

1 Sketch the graph of the following, showing the x- and y-intercepts.

a $x + y = 2$ **b** $x - y = -2$ **c** $2x + y = 4$
d $3x - y = 9$ **e** $4x - 2y = 8$ **f** $3x + 2y = 6$
g $y - 3x = 12$ **h** $-5y + 2x = -10$ **i** $-x + 7y = 21$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 4b

2 Sketch the graph of the following, showing the x- and y-intercepts.
a $y = 3x + 3$
b $y = 2x + 2$
c $y = x - 5$
d $y = -x - 6$
e $y = -2x - 2$
f $y = -2x + 4$

3 Sketch the graph of each of the following mixed linear relationships.
a $x + 2y = 8$
b $3x - 5y = 15$
c $y = 3x - 6$
d $3y - 4x - 12 = 0$
e $2x - y - 4 = 0$
f $2x - y + 5 = 0$

PROBLEM–SOLVING	4, 5	4, 5	5, 6

4 The distance d metres of a vehicle from an observation point after t seconds is given by the rule $d = 8 - 2t$.
a Find the distance from the observation point initially (at $t = 0$).
b Find after what time t the distance d is equal to 0 (substitute $d = 0$).
c Sketch a graph of d versus t between the d and t intercepts.

5 The height, h metres, of a lift above the ground after t seconds is given by $h = 100 - 8t$.
a How high is the lift initially (at $t = 0$)?
b How long does it take for the lift to reach the ground ($h = 0$)?

6 Find the x- and y-axis intercept coordinates of the graphs with the given rules. Write answers using fractions.
a $3x - 2y = 5$
b $x + 5y = -7$
c $y - 2x = -13$
d $y = -2x - 1$
e $2y = x - 3$
f $-7y = 1 - 3x$

REASONING	7	7, 8	8, 9

7 Use your algebra and fraction skills to help sketch graphs for these relations by finding x- and y-intercepts.
a $\frac{x}{2} + \frac{y}{3} = 1$
b $y = \frac{8 - x}{4}$
c $\frac{y}{2} = \frac{2 - 4x}{8}$

8 Explain why the graph of the equation $ax + by = 0$ must pass through the origin for any values of the constants a and b.

9 Write down the rule for the graph with these axes intercepts. Write the rule in the form $ax + by = d$.
a (0, 4) and (4, 0)
b (0, 2) and (2, 0)
c (0, −3) and (3, 0)
d (0, 1) and (−1, 0)
e $(0, k)$ and $(k, 0)$
f $(0, -k)$ and $(-k, 0)$

ENRICHMENT: Intercept families	–	–	10

10 Find the coordinates of the x- and y-intercepts in terms of the constants a, b and c for these relationships.
a $ax + by = c$
b $y = \frac{a}{b}x + c$
c $\frac{ax - by}{c} = 1$
d $ay - bx = c$
e $ay = bx + c$
f $a(x + y) = bc$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4C Lines with one intercept

Learning intentions
- To be able to use equations to graph vertical lines and horizontal lines
- To know that every line of the form $y = mx$ passes through the origin
- To be able to graph lines of the form $y = mx$ using the origin and one other point

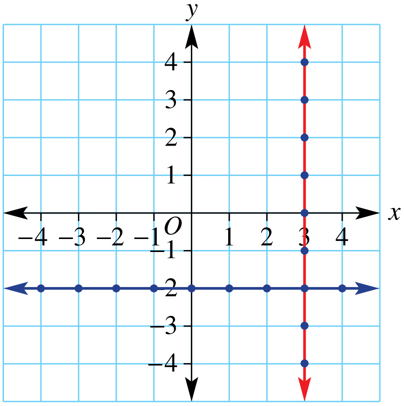
Past, present and future learning
- This section addresses the Stage 5 Core Topic called *Linear relationships A*
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

Lines with one intercept include vertical lines, horizontal lines and lines that pass through the origin.

Linear relationships with graphs through (0, 0) include:
- Fertiliser in kg = usage rate in kg/acre × number of acres;
- annual interest = interest rate p.a. × amount invested;
- weekly pay = pay rate in \$/h × number of hours.

Lesson starter: What rule satisfies all points?

Here is one vertical and one horizontal line.
- For the vertical line shown, write down the coordinates of all the points shown as dots.
- What is always true for each coordinate pair?
- What simple equation describes every point on the line?
- For the horizontal line shown, write down the coordinates of all the points shown as dots.
- What is always true for each coordinate pair?
- What simple equation describes every point on the line?
- Where do the two lines intersect?

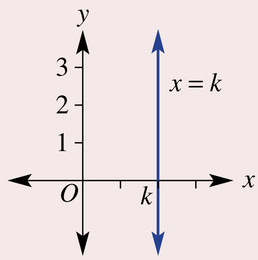


ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

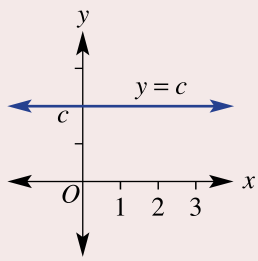
■ Vertical line: $x = k$

- Parallel to the y-axis
- Equation of the form $x = k$, where k is a constant
- x-intercept is at $(k, 0)$



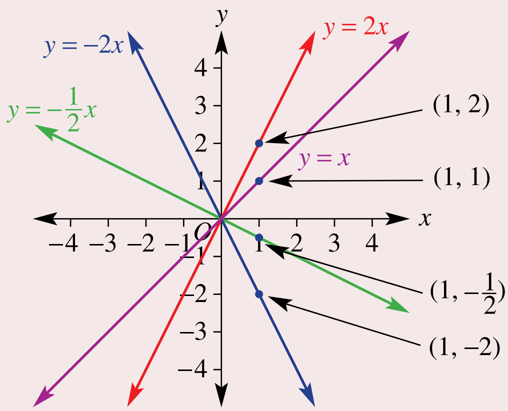
■ Horizontal line: $y = c$

- Parallel to the x-axis
- Equation of the form $y = c$, where c is a constant
- y-intercept is at $(0, c)$



■ Lines through the origin (0, 0): $y = mx$

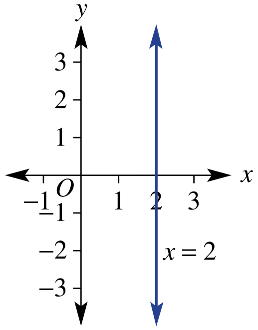
- y-intercept is at (0, 0)
- x-intercept is at (0, 0)
- Substitute $x = 1$ or any other value of x to find a second point

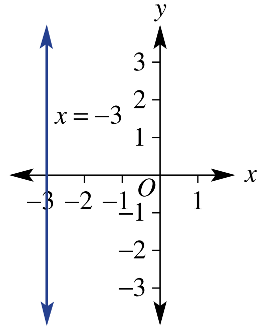


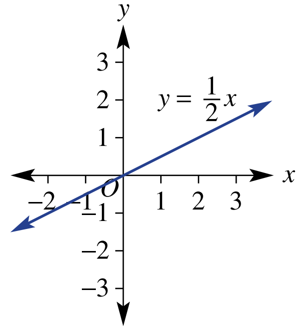
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

BUILDING UNDERSTANDING

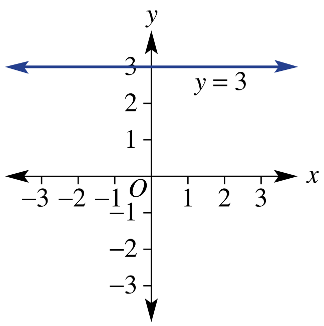
1 State the coordinates of the x-intercept for these graphs.

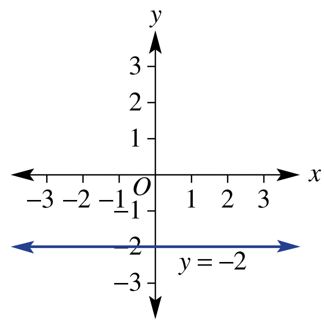
a


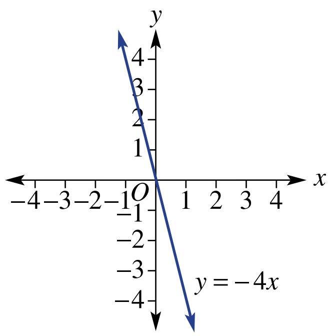
b


c


2 State the coordinates of the y-intercept for these graphs.

a


b


c


3 Find the value of y if $x = 1$ using these rules.

a $y = 5x$ **b** $y = \frac{1}{3}x$ **c** $y = -4x$ **d** $y = -0.1x$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

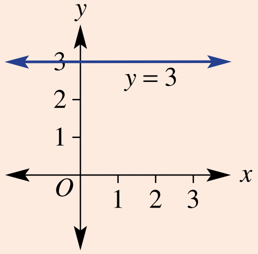
Example 5 Graphing vertical and horizontal lines

Sketch the graph of the following horizontal and vertical lines.

a $y = 3$ **b** $x = -4$

SOLUTION

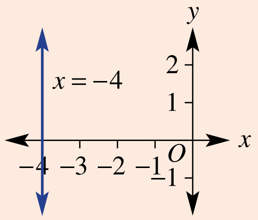
a



EXPLANATION

y-intercept is at $(0, 3)$.
Sketch a horizontal line through all points where $y = 3$.

b



x-intercept is at $(-4, 0)$.
Sketch a vertical line through all points where $x = -4$.

Now you try

Sketch the graph of the following horizontal and vertical lines.

a $y = -1$ **b** $x = 3$

Example 6 Sketching lines that pass through the origin

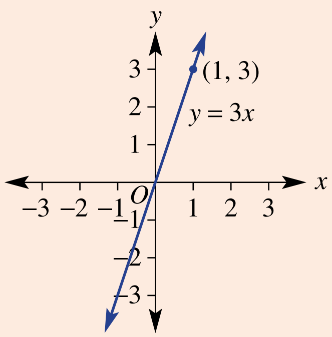
Sketch the graph of $y = 3x$.

SOLUTION

The x- and the y-intercept are both at $(0, 0)$.
Another point (let $x = 1$):

$$y = 3 \times (1)$$
$$y = 3$$

Another point is at $(1, 3)$.



EXPLANATION

The equation is of the form $y = mx$.

As two points are required to generate the straight line, find another point by substituting $x = 1$.
Other x-values could also be used.

Plot and label both points and sketch the graph by joining the points in a straight line.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Now you try

Sketch the graph of $y = -2x$.

Exercise 4C

FLUENCY	1–3(½)	1–4(½)	1–4(½)

Example 5

1 Sketch the graph of the following horizontal and vertical lines.

a $x = 2$ **b** $x = 5$ **c** $y = 4$ **d** $y = 1$
e $x = -3$ **f** $x = -2$ **g** $y = -1$ **h** $y = -3$

Example 6

2 Sketch the graph of the following linear relationships that pass through the origin.

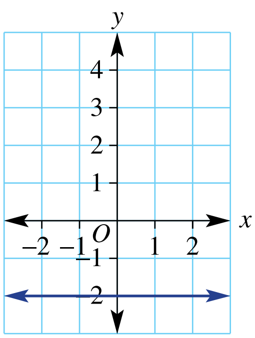
a $y = 2x$ **b** $y = 5x$ **c** $y = 4x$ **d** $y = x$
e $y = -4x$ **f** $y = -3x$ **g** $y = -2x$ **h** $y = -x$

3 Sketch the graphs of these special lines all on the same set of axes and label with their equations.

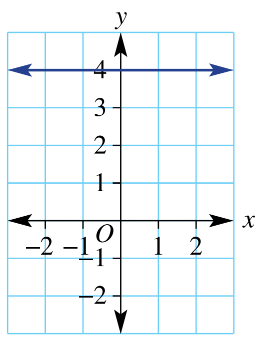
a $x = -2$ **b** $y = -3$ **c** $y = 2$ **d** $x = 4$
e $y = 3x$ **f** $y = -\frac{1}{2}x$ **g** $y = -1.5x$ **h** $x = 0.5$
i $x = 0$ **j** $y = 0$ **k** $y = 2x$ **l** $y = 1.5x$

4 Give the equation of each of the following graphs.

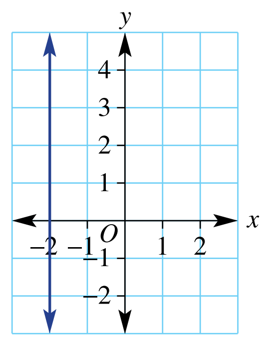
a



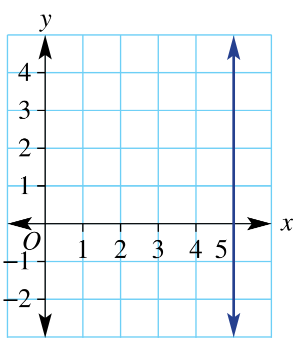
b



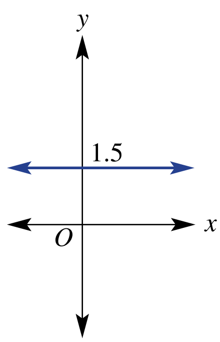
c



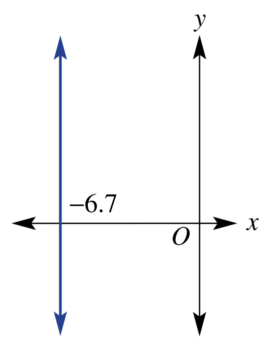
d



e



f



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

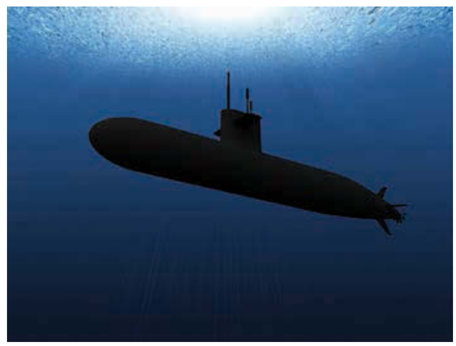
PROBLEM–SOLVING	5, 6	5, 6, 7(½), 8	5(½), 7(½), 8, 9

5 Find the equation of the straight line that is:

a parallel to the x-axis and passes through the point (1, 3)
b parallel to the y-axis and passes through the point (5, 4)
c parallel to the y-axis and passes through the point $(-2, 4)$
d parallel to the x-axis and passes through the point (0, 0).

6 If the surface of the sea is represented by the x-axis, state the equation of the following paths.

a A plane flies horizontally at 250 m above sea level. One unit is 1 metre.
b A submarine travels horizontally 45 m below sea level. One unit is 1 metre.



7 The graphs of these pairs of equations intersect at a point. Find the coordinates of the point.

a $x = 1, y = 2$
b $x = -3, y = 5$
c $x = 0, y = -4$
d $x = 4, y = 0$
e $y = -6x, x = 0$
f $y = 3x, x = 1$
g $y = -9x, x = 3$
h $y = 8x, y = 40$

8 Find the area of the rectangle contained within the following four lines.

a $x = 1, x = -2, y = -3, y = 2$
b $x = 0, x = 17, y = -5, y = -1$

9 The lines $x = -1$, $x = 3$ and $y = -2$ form three sides of a rectangle. Find the possible equation of the fourth line if:

a the area of the rectangle is:
  i 12 square units
  ii 8 square units
  iii 22 square units

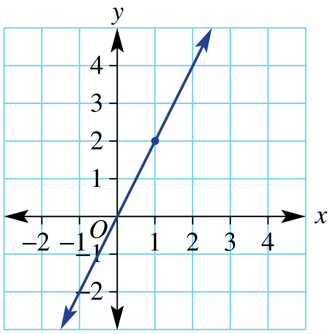
b the perimeter of the rectangle is:
  i 14 units
  ii 26 units
  iii 31 units

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

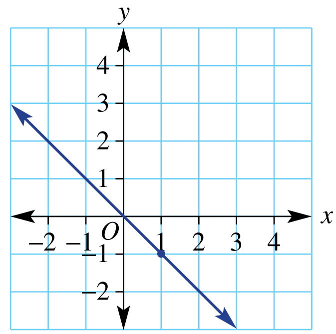
REASONING	10	10–11(½)	10–12(½)

10 The rules of the following graphs are of the form $y = mx$. Use the points marked with a dot to find m and hence state the equation.

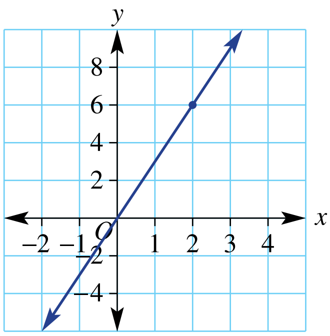
a



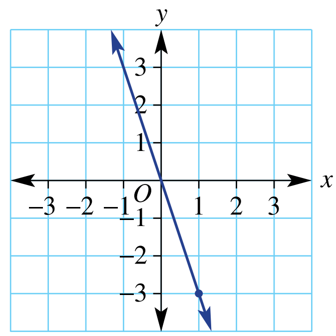
b



c



d



11 Find the equation of the line that passes through the origin and the given point.

a $(1, 3)$ **b** $(1, 4)$ **c** $(1, -5)$ **d** $(1, -2)$

12 Sketch the graph of each of the following by first making y or x the subject.

a $y - 8 = 0$
b $x + 5 = 0$
c $x + \frac{1}{2} = 0$
d $y - 0.6 = 0$
e $y + 3x = 0$
f $y - 5x = 0$
g $2y - 8x = 0$
h $5y + 7x = 0$

ENRICHMENT: Trisection	–	–	13–15

13 A vertical line, horizontal line and another line that passes through the origin all intersect at $(-1, -5)$. What are the equations of the three lines?

14 The lines $y = c$, $x = k$ and $y = mx$ all intersect at one point.

a State the coordinates of the intersection point.
b Find m in terms of c and k.

15 The area of a triangle formed by $x = 4$, $y = -2$ and $y = mx$ is 16 square units. Find the value of m given $m > 0$.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4D Gradient

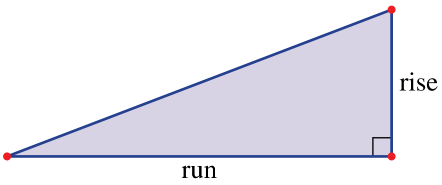
Learning intentions

- To understand what is meant by the gradient of a line
- To know that the gradient of a line can be positive, negative, zero or undefined
- To be able to find the gradient of a line from a graph or using two given points

Past, present and future learning

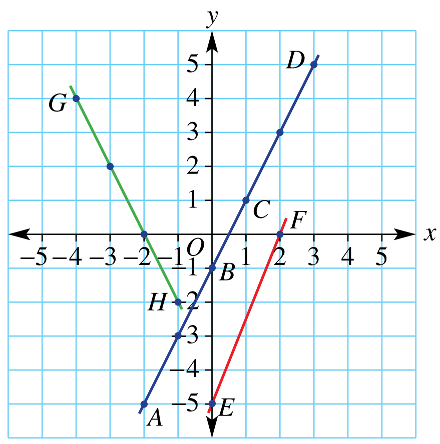
- This section addresses the Stage 5 Core Topic called Linear relationships A
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

The gradient of a line is a measure of its slope. It is a number that describes the steepness of a line and is calculated by considering how far a line rises or falls between two points within a given horizontal distance. The horizontal distance between two points is called the *run* and the vertical distance is called the *rise*.



Lesson starter: Which line is the steepest?

The three lines shown below right connect the points *A*–*H*.



- Calculate the rise and run (working from left to right) and also the fraction $\frac{\text{rise}}{\text{run}}$ for these segments.
  a *AB* **b** *BC* **c** *BD* **d** *EF* **e** *GH*
- What do you notice about the fractions $\frac{\text{rise}}{\text{run}}$ for parts **a**, **b** and **c**?
- How does the $\frac{\text{rise}}{\text{run}}$ for *EF* compare with the $\frac{\text{rise}}{\text{run}}$ for parts **a**, **b** and **c**? Which of the two lines is the steeper?
- Your $\frac{\text{rise}}{\text{run}}$ for *GH* should be negative. Why is this the case?
- Discuss whether or not *GH* is steeper than *AD*.

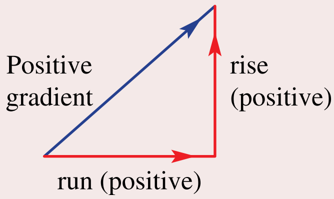
Use computer software (interactive geometry) to produce a set of axes and grid.

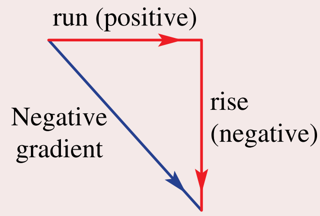
- Construct a line segment with endpoints on the grid. Show the coordinates of the endpoints.
- Calculate the rise (vertical distance between the endpoints) and the run (horizontal distance between the endpoints).
- Calculate the gradient as the *rise* divided by the *run*.
- Now drag the endpoints and explore the effect on the gradient.
- Can you drag the endpoints but retain the same gradient value? Explain why this is possible.
- Can you drag the endpoints so that the gradient is zero or undefined? Describe how this can be achieved.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

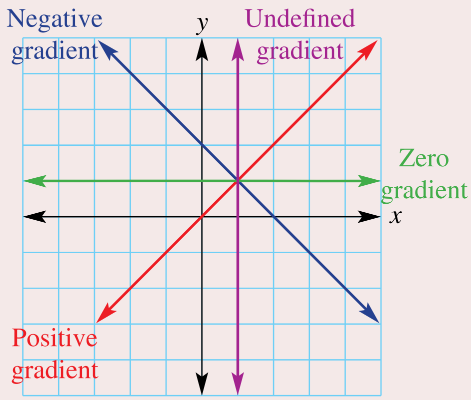
KEY IDEAS

- **Gradient** $(m) = \frac{\text{rise}}{\text{run}}$





- When we work from left to right, the run is always positive.
- The gradient can be positive, negative, zero or undefined.
- A vertical line has an undefined gradient.

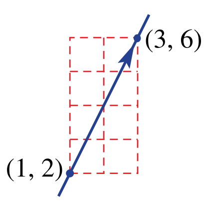


- Gradient is usually expressed as a simplified proper or improper fraction (not a mixed number).

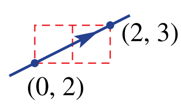
BUILDING UNDERSTANDING

1 Calculate the gradient using $\frac{\text{rise}}{\text{run}}$ for these lines. Remember to give a negative answer if the line is sloping downwards from left to right.

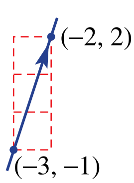
a



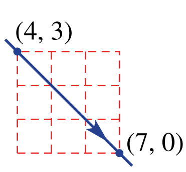
b



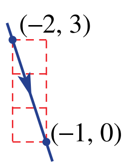
c



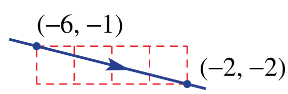
d



e



f



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

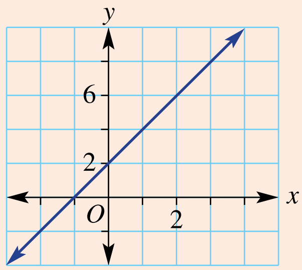
2 Use the words: positive, negative, zero or undefined to complete each sentence.
- **a** The gradient of a horizontal line is ___________.
- **b** The gradient of the line joining (0, 3) with (5, 0) is ___________.
- **c** The gradient of the line joining (−6, 0) with (1, 1) is ___________.
- **d** The gradient of a vertical line is ___________.

3 Use the gradient $m \left(= \frac{\text{rise}}{\text{run}}\right)$ to find the missing value.
- **a** $m = 4$, rise $= 8$, run $= ?$
- **b** $m = 6$, rise $= 3$, run $= ?$
- **c** $m = \frac{3}{2}$, run $= 4$, rise $= ?$
- **d** $m = -\frac{2}{5}$, run $= 15$, rise $= ?$

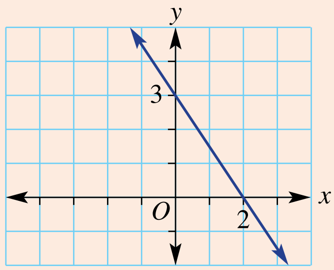
Example 7 Finding the gradient of a line

For each graph, state whether the gradient is positive, negative, zero or undefined, then find the gradient where possible.

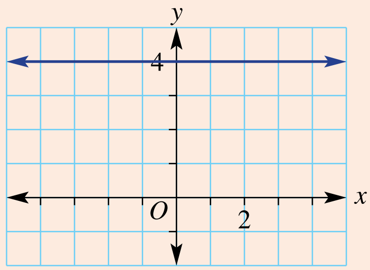
a



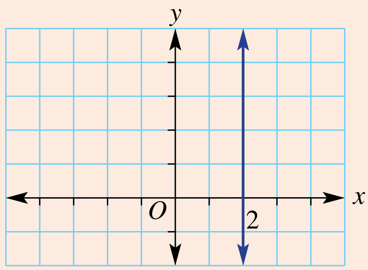
b



c

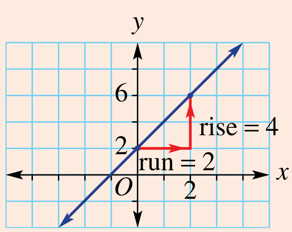


d



SOLUTION

a The gradient is positive.



$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{4}{2} = 2$$

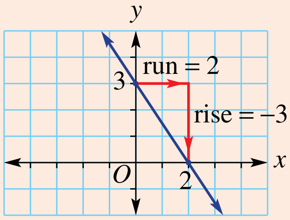
EXPLANATION

By inspection, the gradient will be positive since the graph rises from left to right. Select any two points and create a right-angled triangle to determine the rise and run.

Substitute rise = 4 and run = 2.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

b The gradient is negative.



$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{-3}{2} = -\frac{3}{2}$$

By inspection, the gradient will be negative since y-values decrease from left to right.
Rise $= -3$ and run $= 2$.

c The gradient is 0.

The line is horizontal.

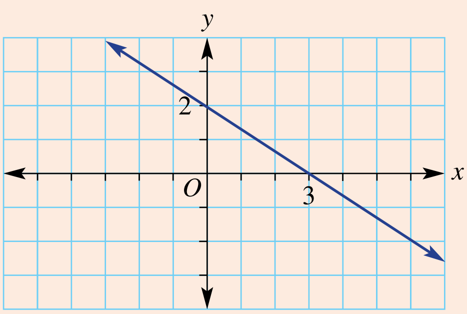
d The gradient is undefined.

The line is vertical.

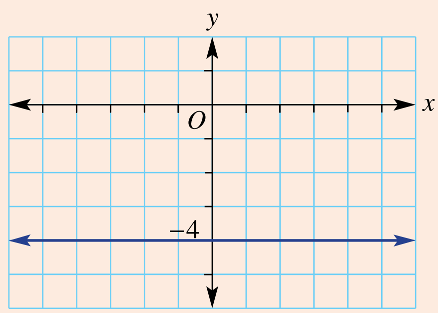
Now you try

For each graph, state whether the gradient is positive, negative, zero or undefined, then find the gradient where possible.

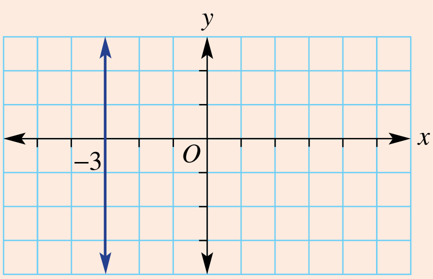
a



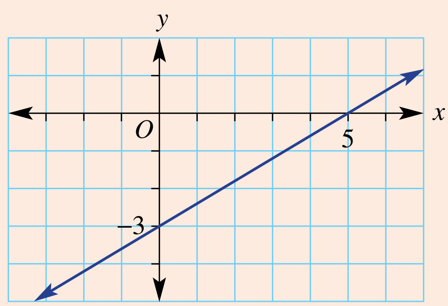
b



c



d



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 8 Finding the gradient between two points

Find the gradient (m) of the line joining the given points.

a $A(3, 4)$ and $B(5, 6)$ **b** $A(-3, 6)$ and $B(1, -3)$

SOLUTION

a $m = \frac{\text{rise}}{\text{run}}$

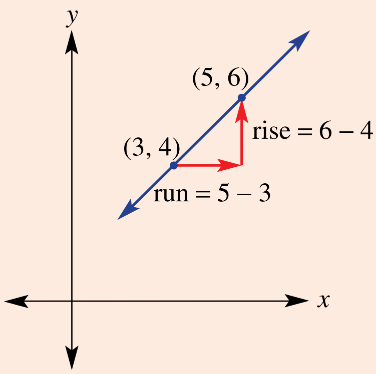
$= \frac{6-4}{5-3}$

$= \frac{2}{2}$

$= 1$

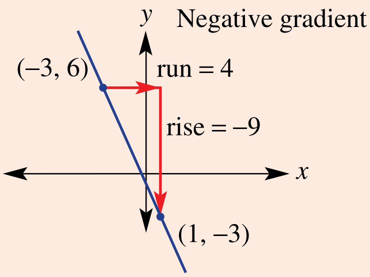
EXPLANATION

Plot points to see a positive gradient and calculate rise and run.



b $m = \frac{\text{rise}}{\text{run}}$

$= \frac{-9}{4}$ or $-\frac{9}{4}$ or -2.25



Now you try

Find the gradient (m) of the line joining the given points.

a $A(2, 6)$ and $B(3, 8)$ **b** $A(-5, 1)$ and $B(2, -3)$

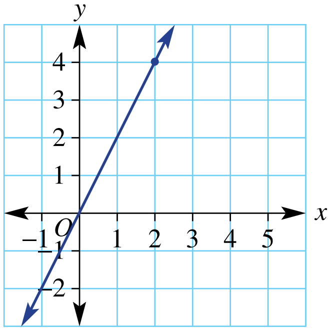
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

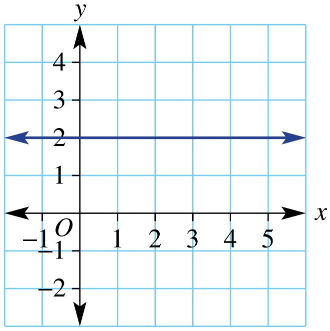
Exercise 4D

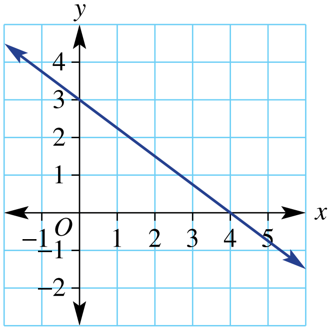
FLUENCY	1, 2(½)	1, 2(½)	1–2(½), 3

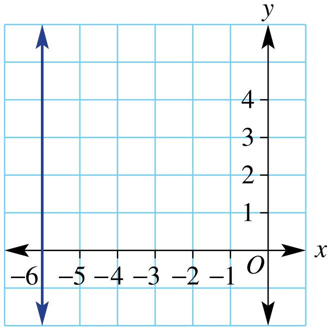
Example 7

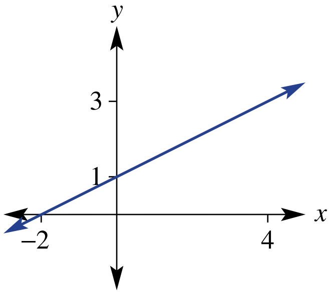
1 For each graph state whether the gradient is positive, negative, zero or undefined, then find the gradient where possible.

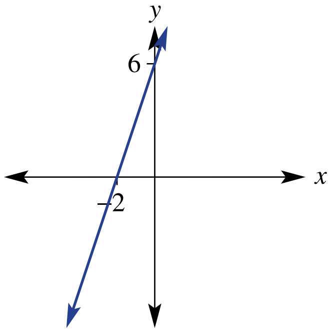
a


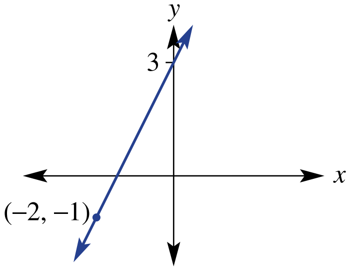
b


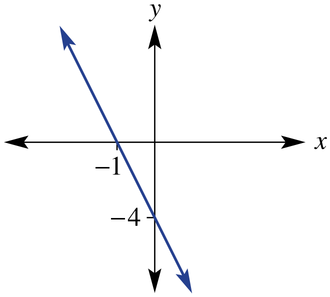
c


d


e


f


g


h


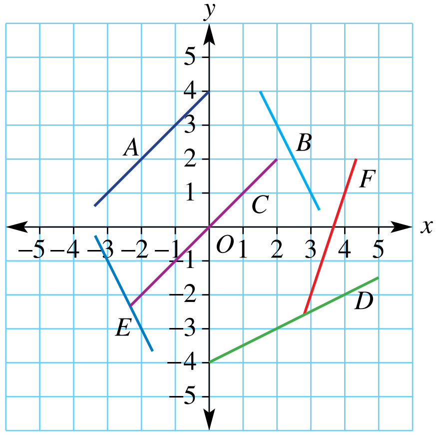
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 8

2 Find the gradient of the lines joining the following pairs of points.

a $A(2, 3)$ and $B(3, 5)$
b $C(2, 1)$ and $D(5, 5)$
c $E(1, 5)$ and $F(2, 7)$
d $G(-2, 6)$ and $H(0, 8)$
e $A(-4, 1)$ and $B(4, -1)$
f $C(-2, 4)$ and $D(1, -2)$
g $E(-3, 4)$ and $F(2, -1)$
h $G(-1, 5)$ and $H(1, 6)$
i $A(-2, 1)$ and $B(-4, -2)$
j $C(3, -4)$ and $D(1, 1)$
k $E(3, 2)$ and $F(0, 1)$
l $G(-1, 1)$ and $H(-3, -4)$

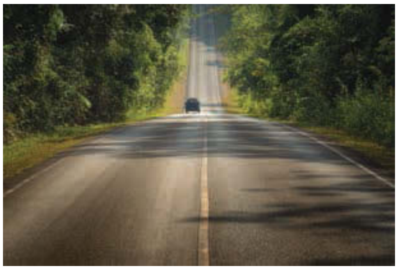
3 Find the gradient of the lines A–F on this graph and grid.



PROBLEM–SOLVING 4, 5 4–6 5–7

4 Find the gradient corresponding to the following slopes.

a A road falls 10 m for every 200 horizontal metres.
b A cliff rises 35 metres for every 2 metres horizontally.
c A plane descends 2 km for every 10 horizontal kilometres.
d A submarine ascends 150 m for every 20 horizontal metres.



5 Find the missing number.

a The gradient joining the points (0, 2) and (1, ?) is 4.
b The gradient joining the points (?, 5) and (1, 9) is 2.
c The gradient joining the points (−3, ?) and (0, 1) is −1.
d The gradient joining the points (−4, −2) and (?, −12) is −4.

6 A train climbs a slope with gradient 0.05. How far horizontally has the train travelled after rising 15 metres?

7 Complete this table showing the gradient, x-intercept and y-intercept for straight lines.

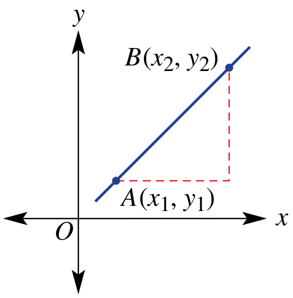
	A	B	C	D	E	F
Gradient	3	−1	$\frac{1}{2}$	$-\frac{2}{3}$	0.4	−1.25
***x*-intercept**	−3			6	1	
***y*-intercept**		−4	$\frac{1}{2}$			3

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

REASONING	8	8, 9	8, 9

8 Give a reason why a line with gradient $\frac{7}{11}$ is steeper than a line with gradient $\frac{3}{5}$.

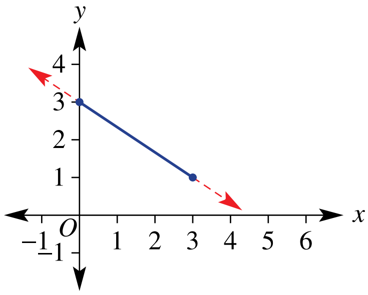
9 The two points A and B shown here have coordinates (x_1, y_1) and (x_2, y_2).



- **a** Write a rule for the run using x_1 and x_2.
- **b** Write a rule for the rise using y_1 and y_2.
- **c** Write a rule for the gradient m using x_1, x_2, y_1 and y_2.
- **d** Use your rule to find the gradient between these pairs of points.
  - **i** (1, 1) and (3, 4)
  - **ii** (0, 2) and (4, 7)
  - **iii** (−1, 2) and (2, −3)
  - **iv** (−4, −6) and (−1, −2)
- **e** Does your rule work for points that include negative coordinates? Explain why.

ENRICHMENT: Where does it hit?	–	–	10

10 The line here has gradient $-\frac{2}{3}$ which means that it falls 2 units for every 3 across. The y-intercept is at (0, 3).



- **a** Use the gradient to find the y-coordinate on the line where:
  - **i** $x = 6$
  - **ii** $x = 9$
- **b** What will be the x-intercept coordinates?
- **c** What would be the x-intercept coordinates if the gradient was changed to the following?
  - **i** $m = -\frac{1}{2}$
  - **ii** $m = -\frac{5}{4}$
  - **iii** $m = -\frac{7}{3}$
  - **iv** $m = -\frac{2}{5}$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4E Gradient and direct proportion

Learning intentions
- To understand that gradient is the rate of change of one variable with respect to another
- To understand what it means for two variables to be directly proportional
- To be able to form and work with equations involving direct variation

Past, present and future learning
- This section addresses the Stage 5 Core Topics called Linear relationships A and Variation A
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

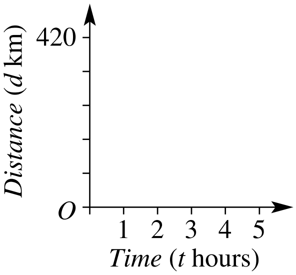
The connection between gradient, rate problems and direct proportion can be illustrated by the use of linear rules and graphs. If two variables are directly related, then the rate of change of one variable with respect to the other is constant. This implies that the rule linking the two variables is linear and can be represented as a straight line graph passing through the origin. The amount of water spraying from a sprinkler, for example, is directly proportional to the time since the sprinkler was turned on. The gradient of the graph of *water volume* versus *time* will equal the rate at which water is spraying from the sprinkler.

A bicycle's speed, v m/min, is in direct proportion to the number of pedal revolutions, P. For example, $v = 6P$ m/min for a wheel of circumference 2 m and gear ratio 3 : 1. If $P = 90$, $v = 540$ m/min, which is a speed of 32.4 km/h.

Lesson starter: Average speed

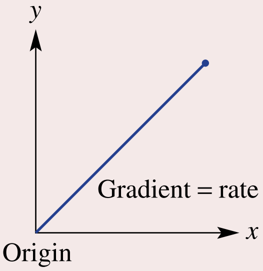
Over 5 hours, Sandy travels 420 km.

- What is Sandy's average speed for the trip?
- Is speed a rate? Discuss.
- Draw a graph of distance versus time, assuming a constant speed.
- Where does your graph intersect the axes and why?
- Find the gradient of your graph. What do you notice?
- Find a rule linking distance (d) and time (t).



KEY IDEAS

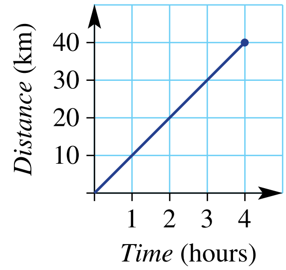
- **Direct variation** occurs when two variables are directly proportional.
  In this situation:
  - the rate of change of one variable with respect to the other is constant
  - the graph is a straight line passing through the origin
  - the rule is of the form $y = mx$ (or $y = kx$)
  - the gradient (m) of the graph equals the rate of change of y with respect to x
  - the letter m (or k) is called the constant of proportionality.



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

BUILDING UNDERSTANDING

1 This graph shows how far a cyclist travels over 4 hours.



a State how far the cyclist has travelled after:
  - **i** 1 hour
  - **ii** 2 hours
  - **iii** 3 hours.

b State the speed of the cyclist (rate of change of distance over time).

c Find the gradient of the graph.

d What do you notice about your answers from parts **b** and **c**?

2 The rule linking the height of a plant and time is given by $h = 5t$ where h is in millimetres and t is in days.

a Find the height of the plant after 3 days.

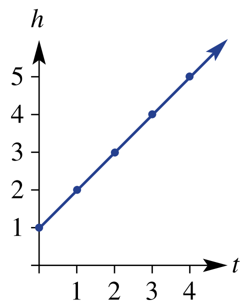
b Find the time for the plant to reach:
  - **i** 30 mm
  - **ii** 10 cm.

c State the missing values for h in this table.

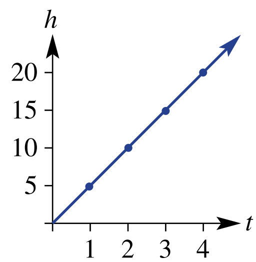
t	0	1	2	3	4
h					

d Choose the correct graph that matches the information above.

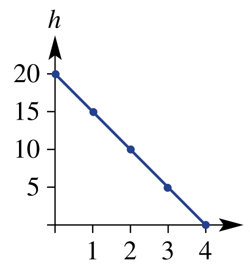
A



B



C



e Find the gradient of the graph.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 9 Exploring direct proportion

Water is poured into an empty tank at a constant rate. It takes 3 hours to fill the tank with 6000 litres.

a What is the rate at which water is poured into the tank?

b Draw a graph of volume (V litres) vs time (t hours) using $0 \leqslant t \leqslant 3$.

c Find:
- **i** the gradient of your graph
- **ii** the rule for V.

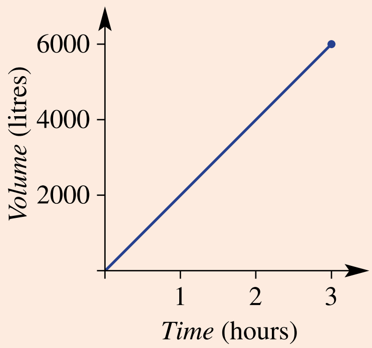
d Use your rule to find:
- **i** the volume after 1.5 hours
- **ii** the time to fill 5000 litres.

SOLUTION

a 6000 L in 3 hours = 2000 L/h

Explanation: 6000 L per 3 hours = 2000 L per 1 hour

b



Explanation: Plot the two endpoints (0, 0) and (3, 6000) then join with a straight line.

c **i** gradient $= \dfrac{6000}{3} = 2000$

Explanation: The gradient is the same as the rate.

ii $V = 2000t$

Explanation: 2000 L are filled for each hour.

d **i**
$$\begin{aligned} V &= 2000t \\ &= 2000 \times (1.5) \\ &= 3000 \text{ litres} \end{aligned}$$

Explanation: Substitute $t = 1.5$ into your rule.

ii
$$\begin{aligned} V &= 2000t \\ 5000 &= 2000t \\ 2.5 &= t \end{aligned}$$

$\therefore$ it takes $2\frac{1}{2}$ hours.

Explanation: Substitute $V = 5000$ into the rule and solve for t.

Now you try

On a long-distance journey a car travels 450 km in 5 hours.

a What is the rate of change of distance over time (i.e. speed)?

b Draw a graph of distance (d km) versus time (t hours) using $0 \leqslant t \leqslant 5$.

c Find:
- **i** the gradient of your graph
- **ii** the rule for d.

d Use your rule to find:
- **i** the distance after 3.5 hours
- **ii** the time to travel 135 km.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Exercise 4E

FLUENCY	1–3	1–4	2–4

Example 9

1 A 300 litre fish tank takes 3 hours to fill from a hose.

a What is the rate at which water is poured into the tank?

b Draw a graph of volume (V litres) vs time (t hours) using $0 \leqslant t \leqslant 3$.

c Find:

i the gradient of your graph

ii the rule for V.

d Use your rule to find:

i the volume after 1.5 hours

ii the time to fill 200 litres.

2 A solar-powered car travels 100 km in 4 hours.

a What is the rate of change of distance over time (i.e. speed)?

b Draw a graph of distance (d km) versus time (t hours) using $0 \leqslant t \leqslant 4$.

c Find:

i the gradient of your graph

ii the rule for d.

d Use your rule to find:

i the distance after 2.5 hours

ii the time to travel 40 km.

3 Write down a rule linking the given variables.

a I travel 600 km in 12 hours. Use d for distance in km and t for time in hours.

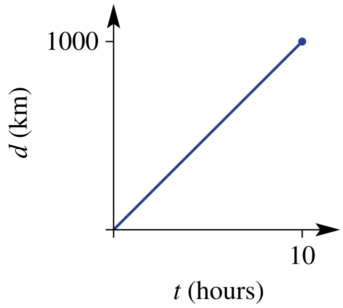
b A calf grows 12 cm in 6 months. Use g for growth height in cm and t for time in months.

c The cost of petrol is \$100 for 80 litres. Use \$$C$ for cost and n for the number of litres.

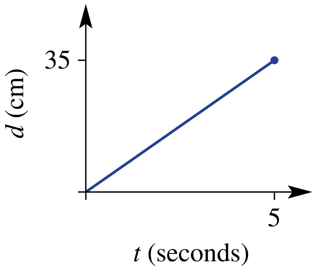
d The profit is \$10 000 for 500 tonnes. Use \$$P$ for profit and t for the number of tonnes.

4 Use the gradient to find the rate of change of distance over time (speed) for these graphs. Use the units given on each graph.

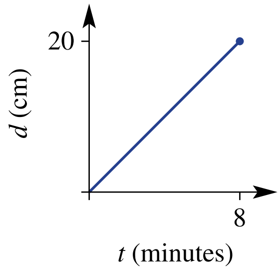
a



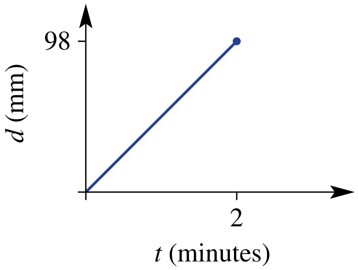
b



c



d



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

PROBLEM–SOLVING	5, 6	5–7	7, 8

5 The trip computer for a car shows that the fuel economy for a trip is 8.5 L per 100 km.
 a How many litres would be used for 120 km?
 b How many litres would be used for 850 km?
 c How many kilometres could be travelled if the capacity of the car's petrol tank was 68 L?

6 Who is travelling the fastest?
- Mick runs 120 m in 20 seconds.
- Sally rides 700 m in 1 minute.
- Udhav jogs 2000 m in 5 minutes.

7 Which animal is travelling the slowest?
- A leopard runs 200 m in 15 seconds.
- A jaguar runs 2.5 km in 3 minutes.
- A panther runs 60 km in 1.2 hours.

8 An investment fund starts at \$0 and grows at a rate of \$100 per month. Another fund starts at \$4000 and reduces by \$720 per year. After how long will the funds have the same amount of money?

REASONING	9	9, 10	10–12

9 The circumference of a circle (given by $C = 2\pi r$) is directly proportional to its radius.
 a Find the circumference of a circle with the given radius. Give an exact answer, e.g. 6π.
  i $r = 0$ ii $r = 2$ iii $r = 6$
 b Draw a graph of C against r for $0 \leqslant r \leqslant 6$. Use exact values for C.
 c Find the gradient of your graph. What do you notice?

10 Is the area of a circle directly proportional to its radius? Give a reason.

11 The base length of a triangle is 4 cm but its height h cm is variable.
 a Write a rule for the area of this triangle.
 b What is the rate at which the area changes with respect to height h?

12 Over a given time interval, is the speed of an object directly proportional to the distance travelled? Give a rule for speed (s) in terms of distance (d) if the time taken is 5 hours.

ENRICHMENT: Rate challenge	–	–	13, 14

13 Hose A can fill a bucket in 2 minutes and hose B can fill the same bucket in 3 minutes. How long would it take to fill the bucket if both hoses were used at the same time?

14 A river is flowing downstream at a rate of 2 km/h. Murray can swim at a rate of 3 km/h. Murray jumps in and swims downstream for a certain distance then turns around and swims upstream back to the start. In total it takes 30 minutes. How far did Murray swim downstream?

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4F Gradient–intercept form

Learning intentions

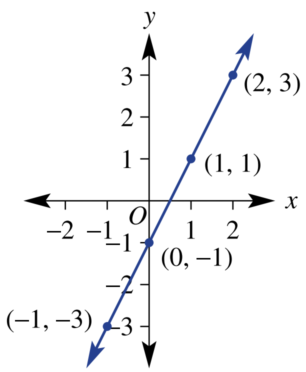
- To know that $y = mx + c$ is the gradient–intercept form of a straight line
- To be able to rearrange a linear equation into gradient–intercept form
- To be able to determine the gradient and y-intercept of a line from the equation
- To be able to sketch a graph using the y-intercept and the gradient

Past, present and future learning

- This section addresses the Stage 5 Core Topics called Linear relationships A and B
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

Shown here is the graph of the rule $y = 2x - 1$. It shows a gradient of 2 and a y-intercept at $(0, -1)$. The fact that these two numbers use numbers in the rule is no coincidence. This is why rules written in this form are called gradient–intercept form. Here are other examples of rules in this form:

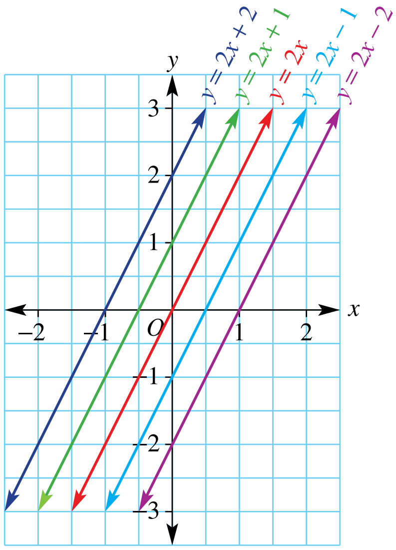
$$y = -5x + 2,\ y = \frac{1}{2}x - 0.5 \text{ and } y = \frac{x}{5} + 20$$



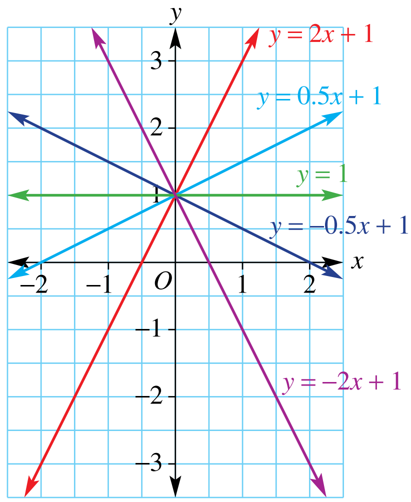
Lesson starter: Family traits

The graph of a linear relation can be sketched easily if you know the gradient and the y-intercept. If one of these is kept constant, we create a family of graphs.

Different y-intercepts and same gradient



Different gradients and same y-intercept



- For the first family, discuss the relationship between the y-intercept and the given rule for each graph.
- For the second family, discuss the relationship between the gradient and the given rule for each graph.

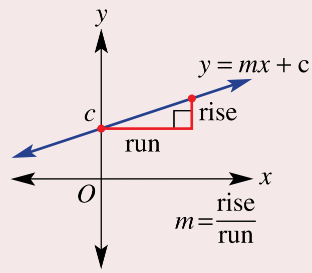
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

- The **gradient–intercept form** of a straight line equation:

  $y = mx + c$ (or $y = mx + b$ depending on preference)

  m = gradient y-intercept = $(0, c)$



- If the y-intercept is $(0, 0)$, the equation becomes $y = mx$ and these graphs will therefore pass through the origin.
- To sketch a graph using the **gradient–intercept method**, locate the y-intercept and use the gradient to find a second point. For example, if $m = \frac{2}{5}$, move 5 across (right) and 2 up.
- The equation of a line can be expressed in several different forms, For example, the line with equation $y = 3x - 1$ can also be written as: $3x - y = 1$ or $3x - y - 1 = 0$ When the RHS is 0, it is said to be in **general form**.

BUILDING UNDERSTANDING

1. State a rule (in gradient–intercept form) for a straight line with the given properties.
   - **a** gradient = 2, y-intercept = $(0, 5)$
   - **b** gradient = -2, y-intercept = $(0, 3)$
   - **c** gradient = -1, y-intercept = $(0, -2)$
   - **d** gradient = $-\frac{1}{2}$, y-intercept = $(0, -10)$
2. Substitute $x = -3$ to find the value of y for these rules.
   - **a** $y = x + 4$
   - **b** $y = 2x + 1$
   - **c** $y = -2x + 3$
3. Rearrange to make y the subject.
   - **a** $y - x = 7$
   - **b** $x + y = 3$
   - **c** $2y - 4x = 10$

Example 10 Stating the gradient and y-intercept

State the gradient and the y-intercept coordinates for the graphs of the following.

a $y = 2x + 1$ **b** $y = -3x$

SOLUTION	EXPLANATION
a $y = 2x + 1$	The rule is given in gradient-intercept form.
gradient = 2	The gradient is the coefficient of x.
y-intercept = $(0, 1)$	The constant term is the y-coordinate of the y-intercept.
b $y = -3x$	
gradient = -3	The gradient is the coefficient of x including the negative sign.
y-intercept = $(0, 0)$	The constant term is not present, so $c = 0$.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Now you try

State the gradient and the y-intercept coordinates for the graphs of the following.

a $y = 5x - 2$ **b** $y = 6x$

Example 11 Rearranging linear equations

Rearrange these linear equations into the form shown in the brackets.

a $4x + 2y = 10$ ($y = mx + c$) **b** $y = 4x - 7$ ($ax + by = d$)

SOLUTION

a
$$\begin{aligned} 4x + 2y &= 10 \\ 2y &= -4x + 10 \\ y &= -2x + 5 \end{aligned}$$

b
$y = 4x - 7$
$y - 4x = -7$
i.e. $-4x + y = -7$
or $4x - y = 7$

EXPLANATION

a Subtract $4x$ from both sides. Here $10 - 4x$ is better written as $-4x + 10$.
Divide both sides by 2. For the right-hand side divide each term by 2.

b Subtract $4x$ from both sides

Multiply both sides by -1 to convert between equivalent forms.

Now you try

Rearrange these linear equations into the form shown in the brackets.

a $10x + 2y = 20$ ($y = mx + c$) **b** $y = -2x + 3$ ($ax + by = d$)

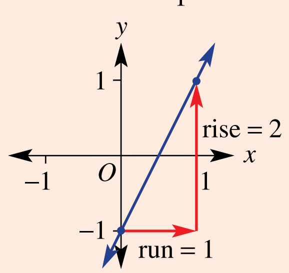
Example 12 Sketching linear graphs using the gradient and y-intercept

Find the value of the gradient and y-intercept for the following and sketch their graphs.

a $y = 2x - 1$ **b** $x + 2y = 6$

SOLUTION

a $y = 2x - 1$
y-intercept $= (0, -1)$
gradient $= 2 = \frac{2}{1}$



EXPLANATION

The rule is in gradient–intercept form so we can read off the gradient and the y-intercept.

Label the y-intercept at $(0, -1)$. From the gradient $\frac{2}{1}$, for every 1 across, move 2 up. From $(0, -1)$ this gives a second point at $(1, 1)$.
Mark and join the points to form a line.

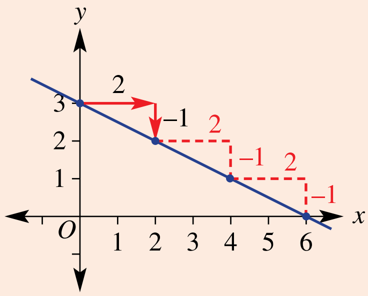
Continued on next page

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

b
$$\begin{aligned} x + 2y &= 6 \\ 2y &= -x + 6 \\ y &= \frac{-x+6}{2} \\ &= -\frac{1}{2}x + 3 \end{aligned}$$

so the y-intercept is $(0, 3)$

$m = -\frac{1}{2} = \frac{-1}{2}$



Rewrite in the form $y = mx + c$ to read off the gradient and y-intercept.

Make y the subject by subtracting x from both sides and then dividing both sides by 2.
$\frac{-x}{2}$ is the same as $-\frac{1}{2}x$ and $6 \div 2 = 3$.

Link the negative sign to the rise (-1) so the run is positive $(+2)$.

Mark the y-intercept at $(0, 3)$ then from this point move 2 right and 1 down to give a second point at $(2, 2)$.

Note that the x-intercept will be at $(6, 0)$. If the gradient is $\frac{-1}{2}$ then a run of 6 gives a fall of 3.

Now you try

Find the value of the gradient and y-intercept for the following and sketch their graphs.

a $y = 3x - 2$ **b** $2x - 5y = 10$

Exercise 4F

FLUENCY	1–3(½)	1–4(½)	1–4(⅓)

Example 10

1 State the gradient and y-intercept coordinates for the graphs of the following.

a $y = 3x + 4$ **b** $y = -5x - 2$ **c** $y = -2x + 3$ **d** $y = \frac{1}{3}x - 4$
e $y = -4x$ **f** $y = 2x$ **g** $y = x$ **h** $y = -0.7x$

Example 11

2 Rearrange these linear equations into the form shown in the brackets.

a $2x + y = 3$ $(y = mx + c)$ **b** $-3x + y = -1$ $(y = mx + c)$
c $6x + 2y = 4$ $(y = mx + c)$ **d** $-3x + 3y = 6$ $(y = mx + c)$
e $y = 2x - 1$ $(ax + by = d)$ **f** $y = -3x + 4$ $(ax + by = d)$
g $3y = x - 1$ (general form) **h** $7y - 2 = 2x$ (general form)

Example 12a

3 Find the gradient and y-intercept for the following and sketch their graphs.

a $y = x - 2$ **b** $y = 2x - 1$ **c** $y = \frac{1}{2}x + 1$ **d** $y = -\frac{1}{2}x + 2$
e $y = -3x + 3$ **f** $y = \frac{3}{2}x + 1$ **g** $y = -\frac{4}{3}x$ **h** $y = \frac{5}{3}x - \frac{1}{3}$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 12b

4 Find the gradient and y-intercept for the following and sketch their graphs. Rearrange each equation first.

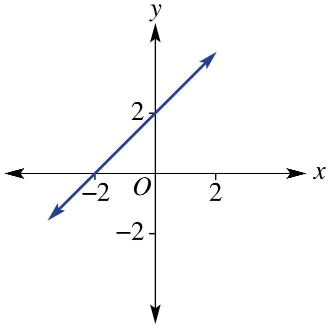
a $x + y = 4$ **b** $x - y = 6$ **c** $x + 2y = 6$
d $x - 2y = 8$ **e** $2x - 3y = 6$ **f** $4x + 3y = 12$
g $x - 3y = -4$ **h** $2x + 3y = 6$ **i** $3x - 4y = 12$
j $x + 4y = 0$ **k** $x - 5y = 0$ **l** $x - 2y = 0$

PROBLEM–SOLVING	5	5, 6(½), 7	6(½), 7

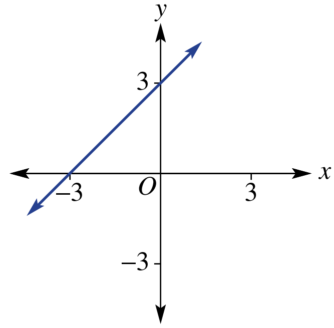
5 Match the following equations to the straight lines shown.

a $y = 3$ **b** $y = -2x - 1$ **c** $y = -2x - 4$
d $y = x + 3$ **e** $x = 2$ **f** $y = x + 2$

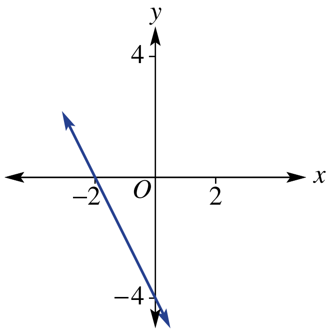
A



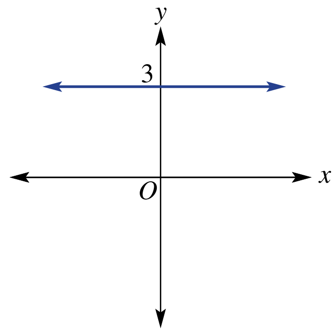
B



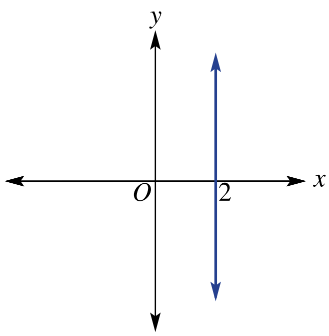
C



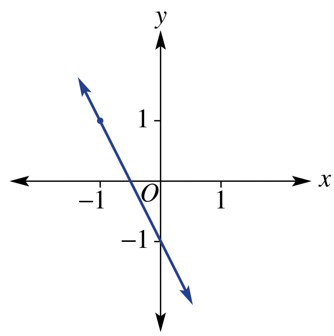
D



E



F



6 Sketch the graph of each of the following linear relationships, by first finding the gradient and y-intercept.

a $5x - 2y = 10$ **b** $y = 6$ **c** $x + y = 0$ **d** $y = 5 - x$
e $y = \frac{x}{2} - 1$ **f** $4y - 3x = 0$ **g** $4x + y - 8 = 0$ **h** $2x + 3y - 6 = 0$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

7 Which of these linear relationships have a gradient of 2 and y-intercept at $(0, -3)$?

A $y = 2(x - 3)$ **B** $y = 3 - 2x$ **C** $y = \dfrac{3 - 2x}{-1}$ **D** $y = 2(x - 1.5)$

E $y = \dfrac{2x - 6}{2}$ **F** $y = \dfrac{4x - 6}{2}$ **G** $2y = 4x - 3$ **H** $-2y = 6 - 4x$

REASONING	8	8, 9	9, 10

8 Jeremy says that the graph of the rule $y = 2(x + 1)$ has gradient 2 and y-intercept $(0, 1)$.

a Explain his error.

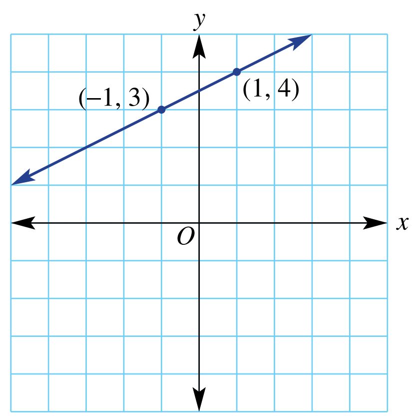
b What can be done to the rule to help show the y-intercept?

9 A horizontal line has gradient 0 and y-intercept at $(0, k)$. Using gradient–intercept form, write the rule for the line.

10 Write the rule $ax + by = d$ in gradient–intercept form. Then state the gradient m and the y-intercept.

ENRICHMENT: The missing y-intercept	–	–	11

11 This graph shows two points $(-1, 3)$ and $(1, 4)$ with a gradient of $\frac{1}{2}$. By considering the gradient, the y-intercept can be calculated to be at $(0, 3.5)$ or $\left(0, \frac{7}{2}\right)$ so $y = \frac{1}{2}x + \frac{7}{2}$.



Use this approach to find the rule of the line passing through these points.

a $(-1, 1)$ and $(1, 5)$

b $(-2, 4)$ and $(2, 0)$

c $(-1, -1)$ and $(2, 4)$

d $(-3, 1)$ and $(2, -1)$

The equation of this straight section of mountain railway could be expressed in the form $y = mx + c$.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

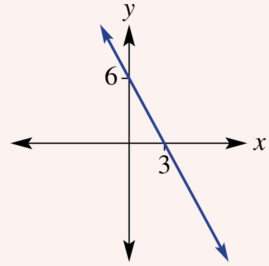
Progress quiz

4A **1** Read off the coordinates of the x- and y-intercepts from this table and graph.

a

x	-2	-1	0	1	2	3
y	-8	-6	-4	-2	0	2

b



4A **2** Decide if the point $(-1, 3)$ is on the line with the given rules.

a $y = 2x - 1$ **b** $y = 2x + 5$

4B **3** Sketch the graph of the following relations, by first finding the x- and y-intercepts.

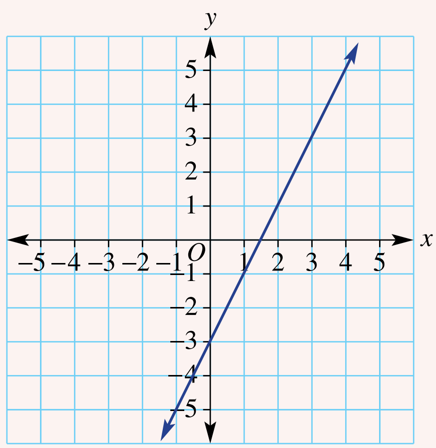
a $3x + 2y = 6$ **b** $y = \frac{3x}{4} + 3$

4C **4** Sketch the graphs of these special lines all on the same set of axes and label with their equations.

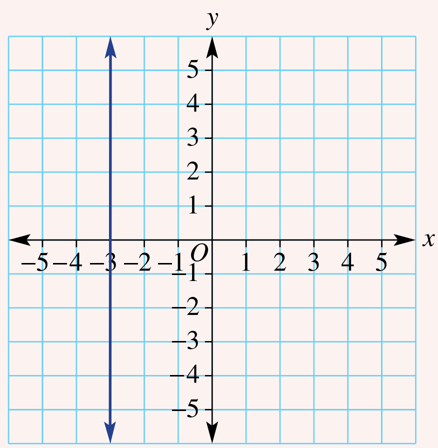
a $y = 2$ **b** $x = 4$ **c** $y = 2x$

4D **5** For each graph state whether the gradient is positive, negative, zero or undefined, then find the gradient where possible.

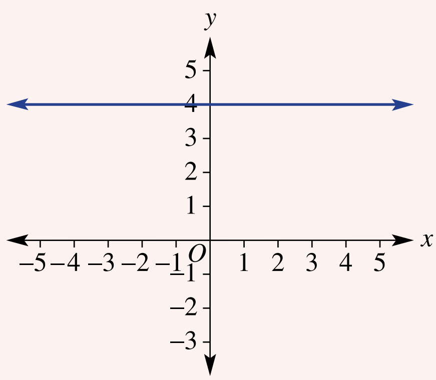
a



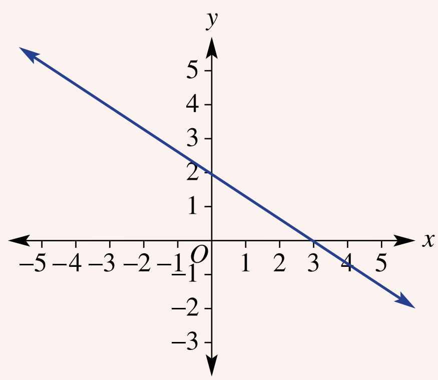
b



c



d



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Progress quiz

4D **6** Find the gradient (m) of the line joining the given points.

a $P(2, 5)$ and $Q(4, 9)$

b $A(-3, 2)$ and $B(1, -10)$

4E **7** An empty tank on a water truck is being filled. It takes 4 hours to fill the tank with 6000 litres.

a What is the rate at which water is poured into the tank?

b Draw a graph of volume (V litres) vs time (t hours) using $0 \leqslant t \leqslant 4$.

c Find:

i the gradient of your graph **ii** the rule for V.

d Use your rule to find:

i the volume after 3 hours **ii** the time to fill 5250 litres.

4F **8** Rearrange these linear equations into the form shown in the brackets.

a $5x + y = 8$ $(y = mx + c)$

b $y = 3x - 5$ $(ax + by = d)$

4F **9** Find the value of the gradient and y-intercept for the following and use these to sketch their graphs.

a $y = 3x - 2$

b $2x + 3y = 9$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4G Finding the equation of a line using $y = mx + c$

Learning intentions

- To understand that all straight lines can be expressed in the form $y = mx + c$
- To be able to find the equation of a line given the y-intercept and the gradient
- To be able to find the equation of a line given the y-intercept and another point
- To be able to find the equation of a line given the gradient and a point

Past, present and future learning

- This section addresses the Stage 5 Core Topics called Linear relationships A and B
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

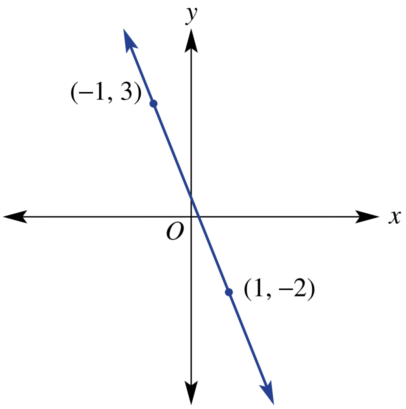
Using the gradient–intercept form, the rule (or equation) of a line can be found by calculating the value of the gradient and the y-intercept. Given a graph, the gradient can be calculated using two points. If the y-intercept is known, then the value of the constant in the rule is obvious. If it is not known, another point can be used to help find its value.

Engineers use linear equations to algebraically model straight supports, such as those in the Madrid airport. Using the slope of the support structure and a location in three dimensions, its linear equation can be found.

Lesson starter: But we don't know the y-intercept!

A line with the rule $y = mx + c$ passes through two points $(-1, 3)$ and $(1, -2)$.

- Using the information given, is it possible to find the value of m? If so, calculate its value.
- The y-intercept is not given on the graph. Discuss what information could be used to find the value of the constant c in the rule. Is there more than one way you can find the y-intercept?
- Write the rule for the line.



KEY IDEAS

- To find the equation of a line in gradient–intercept form $y = mx + c$, you need to find:
  - the value of the gradient (m) using $m = \frac{\text{rise}}{\text{run}}$
  - the value of the constant (c), by observing the y-intercept or by substituting another point.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

BUILDING UNDERSTANDING

1 Substitute the given values of m and c into $y = mx + c$ to find the rule.

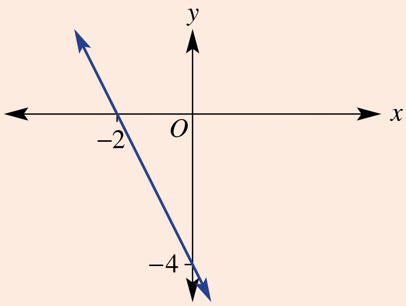
a $m = 2, c = 5$ **b** $m = 4, c = -1$ **c** $m = -2, c = 5$

2 Substitute the point into the given rule and solve to find the value of c. For example, using $(3, 4)$, substitute $x = 3$ and $y = 4$ into the rule.

a $(3, 4), y = x + c$ **b** $(-2, 3), y = 3x + c$ **c** $(3, -1), y = -2x + c$

Example 13 Finding the equation of a line given the y-intercept and another point

Determine the equation of the straight line shown here.



SOLUTION

$$y = mx + c$$

$$m = \frac{\text{rise}}{\text{run}}$$

$$= \frac{-4}{2}$$

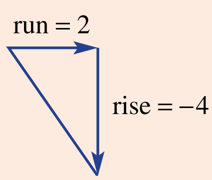
$$= -2$$

y-intercept $= (0, -4)$

$\therefore y = -2x - 4$

EXPLANATION

Write down the general straight line equation.

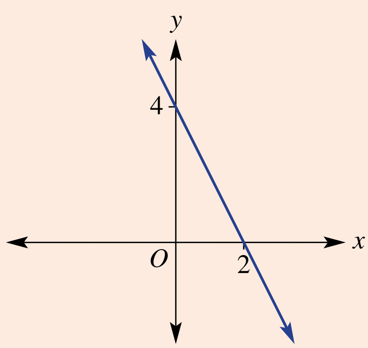


Read from the graph, so $c = -4$.

Substitute $m = -2$ and the value of c into the general equation.

Now you try

Determine the equation of the straight line shown here.



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 14 Finding the equation of a line given the gradient and a point

Find the equation of the line which has a gradient m of $\frac{1}{3}$ and passes through the point (9, 2).

SOLUTION

$$\begin{aligned} y &= mx + c \\ y &= \frac{1}{3}x + c \\ 2 &= \frac{1}{3}(9) + c \\ 2 &= 3 + c \\ -1 &= c \\ \therefore y &= \frac{1}{3}x - 1 \end{aligned}$$

EXPLANATION

Substitute $m = \frac{1}{3}$ into $y = mx + c$.

Since (9, 2) is on the line, it must satisfy the equation $y = \frac{1}{3}x + c$; hence, substitute the point (9, 2) where $x = 9$ and $y = 2$ to find c.

Simplify and solve for c.

Write the equation in the form $y = mx + c$.

Now you try

Find the equation of the line which has a gradient m of $\frac{2}{3}$ and passes through the point $(-3, 4)$.

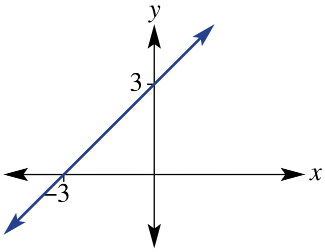
Exercise 4G

FLUENCY	1, 2–3(½)	1–3(½)	1(½), 2–3(⅓)

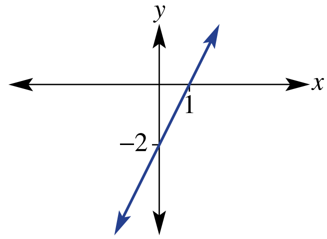
Example 13

1 Determine the equation of the following straight lines.

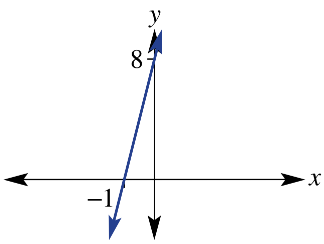
a



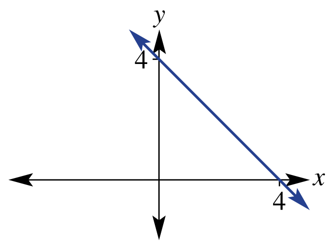
b



c



d



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

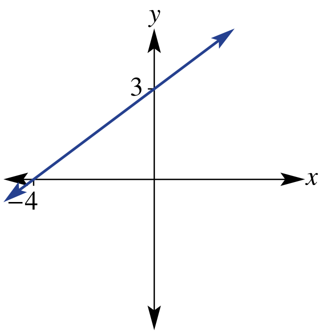
Example 14

2 Find the equation of the line for which the gradient and one point are given.

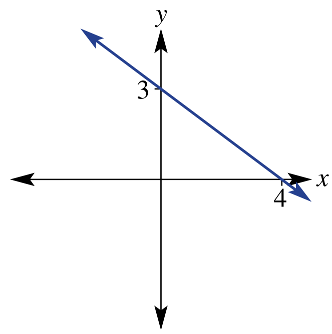
a $m = 3$, point $(1, 8)$
b $m = -2$, point $(2, -5)$
c $m = -3$, point $(2, 2)$
d $m = 1$, point $(1, -2)$
e $m = -3$, point $(-1, 6)$
f $m = 5$, point $(2, 9)$
g $m = -1$, point $(4, 4)$
h $m = -3$, point $(3, -3)$
i $m = -2$, point $(-1, 4)$
j $m = -4$, point $(-2, -1)$

3 Find the equation of these straight lines, which have fractional gradients.

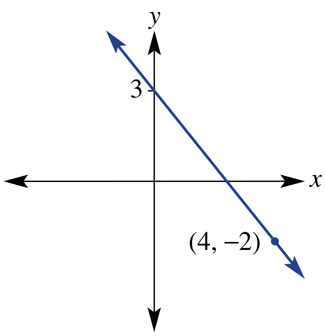
a



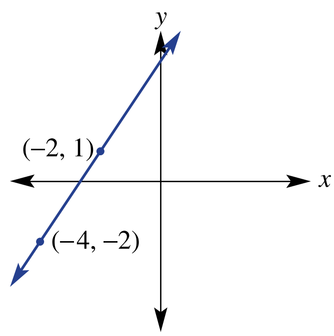
b



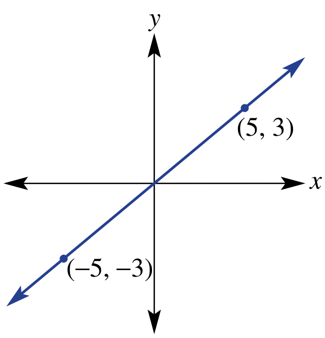
c



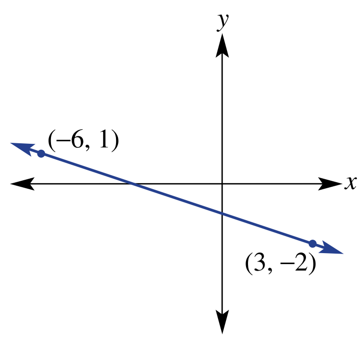
d



e



f



PROBLEM–SOLVING	4(½), 5	4(½), 5–7	4(½), 6–8

4 For the line connecting the following pairs of points:

i find the gradient
ii find the equation.

a $(2, 6)$ and $(4, 10)$
b $(-3, 6)$ and $(5, -2)$
c $(1, 7)$ and $(3, -1)$
d $(-4, -8)$ and $(1, -3)$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

5 A line has gradient -2 and y-intercept (0, 5). Find its x-intercept coordinates.

6 A line passes through the points $(-1, -2)$ and (3, 3). Find its x- and y-intercept coordinates.

7 The tap on a tank has been left on and water is running out. The volume of water in the tank after 1 hour is 100 L and after 5 hours the volume is 20 L. Assuming the relationship is linear, find a rule and then state the initial volume of water in the tank.

8 The coordinates (0, 0) mark the take-off point for a rocket constructed as part of a science class. The positive x direction from (0, 0) is considered to be east.

a Find the equation of the rocket's path if the rocket rises at a rate of 5 m vertically for every 1 m in an easterly direction.

b A second rocket is fired from 2 m vertically above the launch site of the first rocket. It rises at a rate of 13 m for every 2 m in an easterly direction. Find the equation describing its path.

REASONING	9	9(½), 10	10, 11

9 A line has equation $y = mx - 2$. Find the value of m if the line passes through:

a (2, 0) **b** (1, 6) **c** $(-1, 4)$ **d** $(-2, -7)$

10 A line with rule $y = 2x + c$ passes through (1, 5) and (2, 7).

a Find the value of c using the point (1, 5).

b Find the value of c using the point (2, 7).

c Does it matter which point you use? Explain.

11 A line passes through the origin and the point (a, b). Write its equation in terms of a and b.

ENRICHMENT: The general rule	–	–	12

12 To find the equation of a line between two points (x_1, y_1) and (x_2, y_2) some people use the rule:

$$y - y_1 = m(x - x_1) \text{ where } m = \frac{y_2 - y_1}{x_2 - x_1}$$

Use this rule to find the equation of the line passing through these pairs of points. Write your answer in the form $y = mx + c$.

a (1, 2) and (3, 6) **b** (0, 4) and (2, 0) **c** $(-1, 3)$ and (1, 7)

d $(-4, 8)$ and $(2, -1)$ **e** $(-3, -2)$ and (4, 3) **f** $(-2, 5)$ and $(1, -8)$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4H Midpoint and length of a line segment from diagrams

Learning intentions
- To be able to find the midpoint of a line segment given the endpoints
- To be able to use Pythagoras' theorem to find the length of a line segment

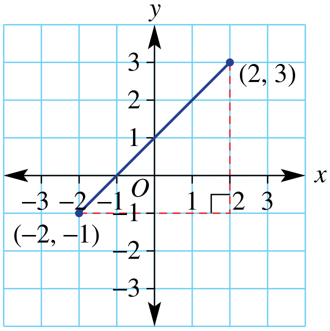
Past, present and future learning
- This section addresses the Stage 5 Core Topic called Linear relationships A
- These concepts will be extended, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

A line segment (or line interval) has a defined length and therefore must have a midpoint. Both the midpoint and length can be found by using the coordinates of the endpoints.

Industrial robots are programmed with the length algorithm to calculate the shortest straight line distance between two points in three dimensions, each located by coordinates (x, y, z).

Lesson starter: Choosing a method

This graph shows a line segment between the points at $(-2, -1)$ and $(2, 3)$.



- What is the horizontal distance between the two points?
- What is the vertical distance between the two points?
- What is the x-coordinate of the point halfway along the line segment?
- What is the y-coordinate of the point halfway along the line segment?
- Discuss and explain a method for finding the midpoint of a line segment.
- Discuss and explain a method for finding the length of a line segment.

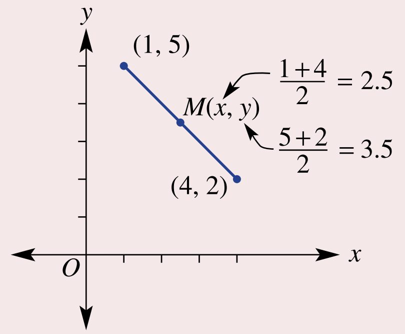
Using graphing software or interactive geometry software, produce a line segment like the one shown above. Label the coordinates of the endpoints and the midpoint. Also find the length of the line segment. Now drag one or both of the endpoints to a new position.

- Describe how the coordinates of the midpoint relate to the coordinates of the endpoints. Is this true for all positions of the endpoints that you choose?
- Now use your software to calculate the vertical distance and the horizontal distance between the two endpoints. Then square these lengths. Describe these squared lengths compared to the square of the length of the line segment. Is this true for all positions of the endpoints that you choose?

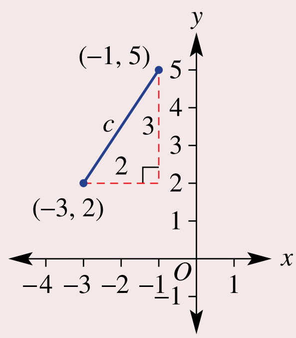
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

- **The midpoint** (M) of a line segment is the halfway point between the two endpoints.
  - The x-coordinate of the midpoint is the average (mean) of the x-coordinates of the two endpoints.
  - The y-coordinate of the midpoint is the average (mean) of the y-coordinates of the two endpoints.
  - $M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$



- The **length of a line segment** (or line interval) is found using Pythagoras' theorem. This gives the distance between any two points.
  - The line segment is the hypotenuse (longest side) of a right-angled triangle.
  - Find the horizontal distance by subtracting the lower x-coordinate from the upper x-coordinate.



$$\begin{aligned}\text{horizontal distance} &= -1 - (-3)\\ &= 2\\ \text{vertical distance} &= 5 - 2\\ &= 3\end{aligned}$$

$$\begin{aligned}c^2 &= 2^2 + 3^2\\ &= 13\\ \therefore c &= \sqrt{13}\end{aligned}$$

  - Find the vertical distance by subtracting the lower y-coordinate from the upper y-coordinate.

BUILDING UNDERSTANDING

1 Find the number that is 'halfway' between these pairs of numbers.

a 1, 7 **b** 5, 11 **c** −2, 4 **d** −6, 0

2 Find the average (mean) of these pairs of numbers.

a 4, 7 **b** 0, 5 **c** −3, 0 **d** −4, −1

3 Evaluate c correct to two decimal places where $c > 0$.

a $c^2 = 1^2 + 2^2$ **b** $c^2 = 5^2 + 7^2$ **c** $c^2 = 10^2 + 2^2$

Example 15 Finding a midpoint of a line segment

Find the midpoint $M(x, y)$ of the line segment joining these pairs of points.

a (1, 0) and (4, 4) **b** (−3, −2) and (5, 3)

SOLUTION

a

$$\begin{aligned}x &= \frac{1+4}{2} = 2.5\\ y &= \frac{0+4}{2} = 2\\ \therefore M &= (2.5, 2)\end{aligned}$$

EXPLANATION

Find the average (mean) of the x-coordinates and the y-coordinates of the two points.

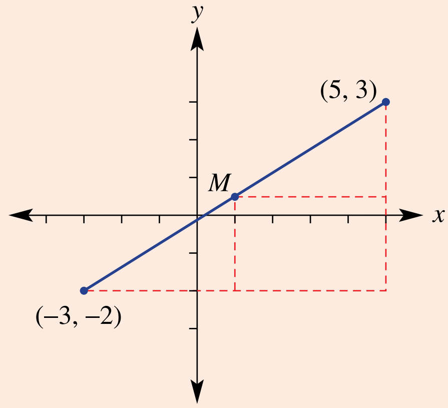
Continued on next page

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

b $x = \dfrac{-3+5}{2} = 1$

$y = \dfrac{-2+3}{2} = 0.5$

$\therefore M = (1, 0.5)$



Now you try

Find the midpoint $M(x, y)$ of the line segment joining these pairs of points.

a (0, 2) and (4, 5)

b (−4, 1) and (2, −4)

Example 16 Finding the length of a line segment

Find the length of the segment joining (−2, 2) and (4, −1), correct to two decimal places.

SOLUTION

Horizontal length $= 4 - (-2)$
$= 6$

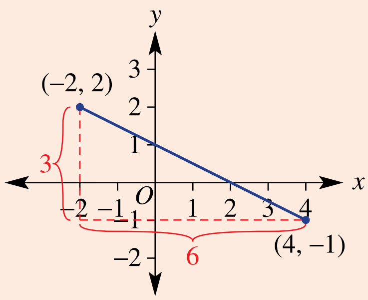
Vertical length $= 2 - (-1)$
$= 3$

$c^2 = 6^2 + 3^2$
$= 45$

$\therefore c = \sqrt{45}$

$\therefore$ length $= 6.71$ units (to 2 d.p.)

EXPLANATION



Apply Pythagoras' theorem $c^2 = a^2 + b^2$.
Round as required.

Now you try

Find the length of the segment joining (−1, 5) and (3, −2), correct to two decimal places.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Exercise 4H

FLUENCY	1–2(½)	1–2(½)	1–2(⅓)

Example 15

1 Find the midpoint $M(x, y)$ of the line segment joining these pairs of points.

a (0, 0) and (6, 6)
b (0, 0) and (4, 4)
c (3, 0) and (5, 2)
d (−2, 0) and (0, 6)
e (−4, −2) and (2, 0)
f (1, 3) and (2, 0)
g (−3, 7) and (4, −1)
h (−2, −4) and (−1, −1)
i (−7, −16) and (1, −1)
j (−4, −3) and (5, −2)

Example 16

2 Find the length of the segment joining these pairs of points correct to two decimal places.

a (1, 1) and (2, 6)
b (1, 2) and (3, 4)
c (0, 2) and (5, 0)
d (−2, 0) and (0, −4)
e (−1, 3) and (2, 1)
f (−2, −2) and (0, 0)
g (−1, 7) and (3, −1)
h (−4, −1) and (2, 3)
i (−3, −4) and (3, −1)

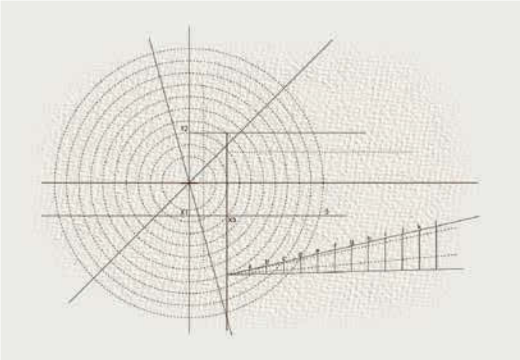
PROBLEM–SOLVING	3, 4	3–5	4–6

3 Find the missing coordinates in this table. M is the midpoint of points A and B.

A	*B*	*M*
(4, 2)		(6, 1)
	(0, −1)	(−3, 2)
	(4, 4)	(−1, 6.5)

4 A circle has centre (2, 1). Find the coordinates of the endpoint of a diameter if the other endpoint has these coordinates.

a (7, 1)
b (3, 6)
c (−4, −0.5)



5 Find the perimeter of these shapes correct to one decimal place.

a A triangle with vertices (−2, 0), (−2, 5) and (1, 3)
b A trapezium with vertices (−6, −2), (1, −2), (0, 4) and (−5, 4)

6 Find the coordinates of the four points which have integer coordinates and are a distance of $\sqrt{5}$ from the point (1, 2). (*Hint*: $5 = 1^2 + 2^2$.)

REASONING	7	7, 8	7, 8

7 A line segment has endpoints (x_1, y_1) and (x_2, y_2) and midpoint $M(x, y)$.

a Write a rule for x, the x-coordinate of the midpoint.
b Write a rule for y, the y-coordinate of the midpoint.
c Test your rule to find the coordinates of M if $x_1 = -3$, $y_1 = 2$, $x_2 = 5$ and $y_2 = -3$.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

8 A line segment has endpoints (x_1, y_1) and (x_2, y_2). Assume $x_2 > x_1$ and $y_2 > y_1$.

a Write a rule for:

i the horizontal distance between the endpoints

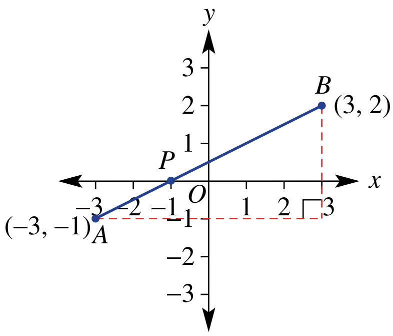
ii the vertical distance between the endpoints

iii the length of the segment.

b Use your rule to show that the length of the segment joining $(-2, 3)$ with $(1, -3)$ is $\sqrt{45}$.

ENRICHMENT: Division by ratio	–	–	9

9 Looking from left to right, this line segment shows the point $P(-1, 0)$, which divides the segment in the ratio 1 : 2.



a What fraction of the horizontal distance between the endpoints is P from A?

b What fraction of the vertical distance between the endpoints is P from A?

c Find the coordinates of point P on the segment AB if it divides the segment in these ratios.

i 2 : 1

ii 1 : 5

iii 5 : 1

d Find the coordinates of point P, which divides the segments with the given endpoints in the ratio 2 : 3.

i $A(-3, -1)$ and $B(2, 4)$

ii $A(-4, 9)$ and $B(1, -1)$

iii $A(-2, -3)$ and $B(4, 0)$

iv $A(-6, -1)$ and $B(3, 8)$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4I Perpendicular lines and parallel lines

Learning intentions
- To know the relationship between the gradients of lines that are either parallel or perpendicular
- To be able to find the equation of a line given a point and a line to which it is parallel or perpendicular
- Past, present and future learning
- This section addresses the Stage 5 Path Topic called Linear relationships C
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

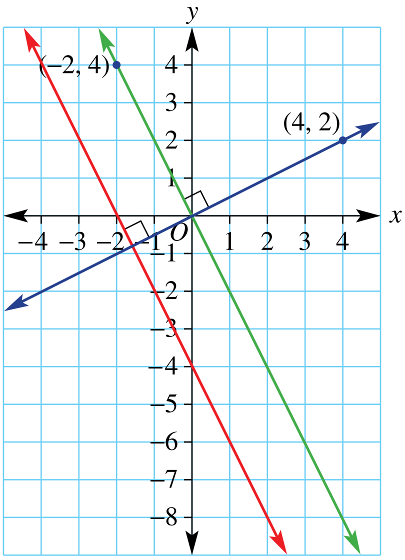
Parallel and perpendicular lines are commonplace in mathematics and in the world around us. Using parallel lines in buildings, for example, ensures that beams or posts point in the same direction. Perpendicular beams help to construct rectangular shapes, which are central in the building of modern structures.

The architect who designed this building has used linear equations to model the sets of parallel lines on its exterior. Shapes with parallel sides are arranged to emphasise the many parallel lines.

Lesson starter: How are they related?

This graph shows a pair of parallel lines and a line perpendicular to the other two. Find the equation of all three lines.

- What do you notice about the equations for the pair of parallel lines?
- What do you notice about the gradient of the line that is perpendicular to the other two lines?
- Write down the equations of three other lines that are parallel to $y = -2x$.
- Write down the equations of three other lines that are perpendicular to $y = -2x$.



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

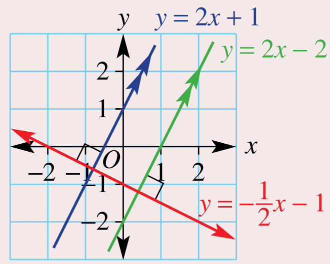
- If two lines are **parallel** then they have the same gradient.
- If two **perpendicular** lines (at right angles) have gradients m_1 and m_2 then:

  $$m_1 \times m_2 = -1 \text{ or } m_2 = -\frac{1}{m_1}$$

  - The **negative reciprocal** of m_1 gives m_2.

    For example: If $m_1 = 4$ then $m_2 = -\frac{1}{4}$.

    If $m_1 = -\frac{2}{3}$ then $m_2 = \frac{-1}{\left(-\frac{2}{3}\right)} = -1 \times \left(-\frac{3}{2}\right) = \frac{3}{2}$.



BUILDING UNDERSTANDING

1 Decide if the pairs of lines with these equations are parallel (have the same gradient).

a $y = 3x - 1$ and $y = 3x + 4$ **b** $y = 7x - 2$ and $y = 2x - 7$

c $y = x + 4$ and $y = x - 3$ **d** $y = \frac{1}{2}x - 1$ and $y = -\frac{1}{2}x + 2$

2 Two perpendicular lines with gradients m_1 and m_2 are such that $m_2 = -\frac{1}{m_1}$. Find m_2 for the given values of m_1.

a $m_1 = 5$ **b** $m_1 = -3$

3 Decide if the pairs of lines with these equations are perpendicular (i.e. whether $m_1 \times m_2 = -1$).

a $y = 4x - 2$ and $y = -\frac{1}{4}x + 3$ **b** $y = 2x - 3$ and $y = \frac{1}{2}x + 4$

c $y = -\frac{1}{2}x + 6$ and $y = -\frac{1}{2}x - 2$ **d** $y = -\frac{1}{5}x + 1$ and $y = 5x + 2$

Example 17 Finding the equation of a parallel line

Find the equation of a line which is parallel to $y = 3x - 1$ and passes through (0, 4).

SOLUTION

$$\begin{aligned} y &= mx + c \\ m &= 3 \\ c &= 4 \\ \therefore y &= 3x + 4 \end{aligned}$$

EXPLANATION

Since it's parallel to $y = 3x - 1$, the gradient is the same so $m = 3$.

The y-intercept is given in the question, so $c = 4$.

Now you try

Find the equation of a line which is parallel to $y = 4x + 3$ and passes through $(0, -2)$.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 18 Finding the equation of a perpendicular line

Find the equation of a line which is perpendicular to the line $y = 2x - 3$ and passes through $(0, -1)$.

SOLUTION

$$y = mx + c$$
$$m = -\frac{1}{2}$$
$$c = -1$$
$$y = -\frac{1}{2}x - 1$$

EXPLANATION

Since it is perpendicular to $y = 2x - 3$,
$m_2 = -\frac{1}{m_1} = -\frac{1}{2}$.

The y-intercept is given.

Now you try

Find the equation of a line which is perpendicular to the line $y = -3x + 1$ and passes through $(0, 4)$.

Exercise 4I

FLUENCY	1, 2(½)	1, 2(½), 3	1, 2(½), 3

Example 17

1 Find the equation of the line that is parallel to the given line and passes through the given point.

a $y = 4x + 2$, $(0, 8)$
b $y = -2x - 3$, $(0, -7)$
c $y = \frac{2}{3}x + 6$, $(0, -5)$
d $y = -\frac{4}{5}x - 3$, $\left(0, \frac{1}{2}\right)$

Example 18

2 Find the equation of the line that is perpendicular to the given line and passes through the given point.

a $y = 3x - 2$, $(0, 3)$
b $y = 5x - 4$, $(0, 7)$
c $y = -2x + 3$, $(0, -4)$
d $y = -x + 7$, $(0, 4)$
e $y = -7x + 2$, $\left(0, -\frac{1}{2}\right)$
f $y = x - \frac{3}{2}$, $\left(0, \frac{5}{4}\right)$

3 a Write the equation of the line parallel to $y = 4$ which passes through these points.
i $(0, 1)$ ii $(-3, -2)$
b Write the equation of the line parallel to $x = -2$ which passes through these points.
i $(3, 0)$ ii $(-3, -3)$
c Write the equation of the line perpendicular to $y = -3$ which passes through these points.
i $(2, 0)$ ii $(0, 0)$
d Write the equation of the line perpendicular to $x = 6$ which passes through these points.
i $(0, 7)$ ii $\left(0, -\frac{1}{2}\right)$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

PROBLEM-SOLVING	4	4, 5	4(½), 5, 6

4 Find the equation of the line which is:

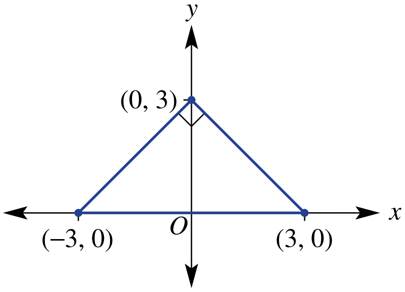
a parallel to the line with equation $y = -3x - 7$ and passes through (3, 0); remember to substitute the point (3, 0) to find the value of the y-intercept

b parallel to the line with equation $y = \frac{1}{2}x + 2$ and passes through (1, 3)

c perpendicular to the line with equation $y = 5x - 4$ and passes through (1, 6)

d perpendicular to the line with equation $y = -x - \frac{1}{2}$ and passes through $(-2, 3)$.

5 A right-angled isosceles triangle has vertices at (0, 3), (3, 0) and $(-3, 0)$. Find the equation of each side.



6 A parallelogram has two side lengths of 5 units. Three of its sides have equations $y = 0$, $y = 2$, $y = 2x$. Find the equation of the fourth side.

REASONING	7	7, 8	7, 8

7 a Using $m_2 = -\frac{1}{m_1}$, find the gradient of a line perpendicular to the line with the given gradient.

i $m_1 = \frac{2}{3}$ ii $m_1 = \frac{1}{5}$ iii $m_1 = -\frac{1}{7}$ iv $m_1 = -\frac{3}{11}$

b If $m_1 = \frac{a}{b}$, find m_2 given $m_1 \times m_2 = -1$.

8 a Find the gradient of a line that is parallel to these lines.

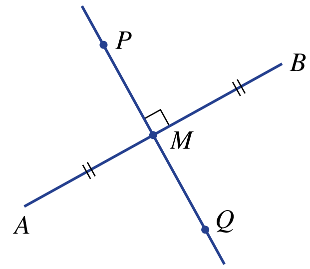
i $2x + 4y = 9$ ii $3x - y = 8$

b Find the gradient of a line that is perpendicular to these lines.

i $5x + 5y = 2$ ii $7x - y = -1$

ENRICHMENT: Perpendicular bisectors	–	–	9(½)

9 If a line segment AB is cut by another line PQ at right angles through the midpoint (M) of AB, then PQ is called the perpendicular bisector. By first finding the midpoint of AB, find the equation of the perpendicular bisector of the segment connecting these points.



a $A(1, 1)$, $B(3, 5)$

b $A(0, 6)$, $B(4, 0)$

c $A(-2, 3)$, $B(6, -1)$

d $A(-6, -1)$, $B(0, 2)$

e $A(-1, 3)$, $B(2, -4)$

f $A(-6, -5)$, $B(4, 7)$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

The following problems will investigate practical situations drawing upon knowledge and skills developed throughout the chapter. In attempting to solve these problems, aim to identify the key information, use diagrams, formulate ideas, apply strategies, make calculations and check and communicate your solutions.

Solar car battery

1 The battery power of two electric cars is analysed and compared by data scientists after testing the cars under average city driving conditions. The average distance covered in 1 hour under such conditions is 40 km.

- The Zet car battery is fully charged with 80 kWh reducing at a rate of 20 kWh per hour.
- The Spark car battery is fully charged with 60 kWh reducing at a rate of 12 kWh per hour.

The data scientists wish to investigate the life of each battery charge and compare the performance of each car's battery within the limits of average city driving conditions.

a Find a rule for the battery power P (kWh) over t hours for:
  i the Zet car
  ii the Spark car.

b Find the battery power remaining (the value of P (kWh)) after 3 hours of average city driving for:
  i the Zet car
  ii the Spark car.

c After how long will there be 40 kWh of power remaining for:
  i the Zet car?
  ii the Spark car?

d Which car's battery has the longer travel time? Give reasons.

e If the cars are driven until their batteries are entirely depleted of power, how far could they travel under average city conditions? Recall that 40 km is covered in one hour.

f After how long will the cars have the same remaining battery power?

g A third car, the Watt, has a battery producing the following statistics:
  - 60 kWh after 1 hour of driving
  - 20 kWh after 4 hours of driving.

  Find a rule for the Watt's remaining battery power P (kWh) after t hours and calculate how far this car can travel on one full charge.

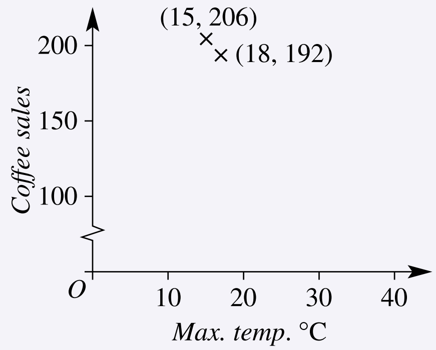
Temperature vs coffee

2 Lydia owns a coffee shop and notices a decline in coffee sales as the maximum daily temperature rises over summer. She collects the following statistics by selecting 5 days within a two-month period.

Data point	1	2	3	4	5
Max. temperature	15	18	27	31	39
Coffee sales	206	192	165	136	110

Lydia wishes to analyse her collected data and see if there is a clear relationship between the maximum temperature and coffee sales. She wants to be better informed, so she can predict income from coffee sales in the future.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.



a Copy and complete this graph for the 5 data points. Two points have been done for you.

b Do you think there is a clear relationship between coffee sales and maximum temperature? Give a reason.

c Using the coordinates of data points 1 and 5, find a rule linking coffee sales, C, and maximum temperature, T.

d Using your rule from part **c**, predict the number of coffee sales for a day with the following maximum temperatures.

   i 25°C **ii** 35°C

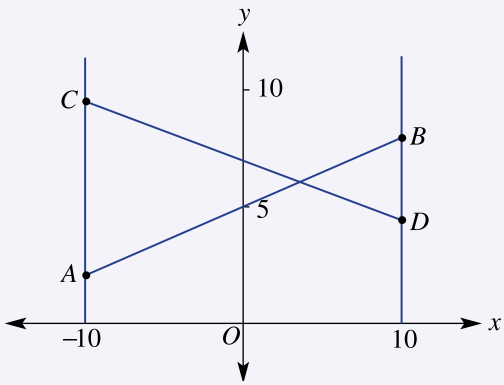
e Using your rule from part **c**, predict the maximum temperature for a day with the following number of coffee sales.

   i 150 **ii** 202

f At what maximum temperature does your rule predict that there will be no coffee sales? Is this realistic?

Rope course

3 Douglas is designing a rope obstacle course between two vertical trees which are 20 m apart. To help analyse his design, he draws the two trees represented by $x = -10$ and $x = 10$ on a Cartesian plane as shown.
Two initial ropes AB and CD are used to start with. Douglas sets the heights of the points A, B, C and D as follows:
$A = 3$ m, $B = 7$ m, $C = 9$ m, $D = 4$ m



Douglas wants to be clear about his design so that he has a good idea about the amount of rope required and the positions of various key points on the obstacle course.

a Find the lengths of the following ropes correct to two decimal places.

   i AB **ii** CD

b Find the equation of the lines which describes the following ropes.

   i AB **ii** CD

c Use an accurate graph to determine the approximate position at which the ropes intersect.

d Douglas decides to place a third rope midway along and perpendicular to rope AB and run it from this position to the left tree.

   i Find an equation for this third rope. **ii** How high up the left tree will it reach?

   iii What will be the length of the rope correct to one decimal place?

e Douglas decides that he wants a fourth rope running from the point D on the right tree to the left tree, joining somewhere between point A and the midpoint of AC. What are the minimum and maximum lengths of rope that he might need? Round to one decimal place where necessary.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4J Linear modelling

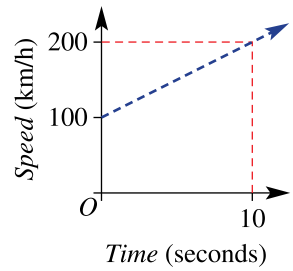
Learning intentions

- To understand that many real-life situations can be modelled using a straight line
- To be able to find the equation linking two variables
- To know where the dependent and independent variable are positioned on a set of axes
- To be able to use an equation or graph to estimate the value of one variable given the other

Past, present and future learning

- This section addresses the Stage 5 Core Topic called Linear relationships A
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

If a relationship between two variables is linear, the graph will be a straight line and the equation linking the two variables can be written in gradient–intercept form. The process of describing and using such line graphs and rules for the relationship between two variables is called linear modelling. A test car, for example, increasing its speed from 100 km/h to 200 km/h in 10 seconds with constant acceleration could be modelled by the rule $s = 10t + 100$. This rule could then be used to calculate the speed at different times in the test run.



A doctor may prescribe 1400 mL of saline solution to be given intravenously over seven hours. The rate of infusion is 200 mL/h, and the remaining volume, V, after t hours has the linear equation: $V = -200t + 1400$.

Lesson starter: The test car

The graph shown above describes the speed of a racing car over a 10 second period.

- Explain why the rule is $s = 10t + 100$.
- Why might negative values of t not be considered for the graph?
- How could you accurately calculate the speed after 6.5 seconds?
- If the car continued to accelerate at the same rate, how could you accurately predict the car's speed after 13.2 seconds?

KEY IDEAS

- Many situations can often be **modelled** by using a linear rule or graph. The key elements of linear modelling include:
  - finding the rule linking the two variables
  - sketching a graph

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

- using the graph or rule to predict or estimate the value of one variable given the other
- finding the rate of change of one variable with respect to the other variable – this is equivalent to finding the gradient.

BUILDING UNDERSTANDING

1 A person gets paid \$50 plus \$20 per hour. Decide which rule describes the relationship between the pay $\$P$ and number of hours worked, n.

A $P = 50 + n$ **B** $P = 50 + 20n$
C $P = 50n + 20$ **D** $P = 20 + 50$

2 The amount of money in a bank account is \$1000 and is increasing by \$100 per month.

a Find the amount of money in the account after:
 i 2 months **ii** 5 months **iii** 12 months

b State a rule for the amount of money A dollars after n months.

3 If $d = 5t - 4$, find:

a d when $t = 10$ **b** d when $t = 1.5$
c t when $d = 6$ **d** t when $d = 11$

Example 19 Applying linear relations

The deal offered by Bikeshare, a bicycle rental company, is a fixed charge of \$20 plus \$5 per hour of use.

a Write a rule for the total cost, $\$C$, of renting a bike for t hours.
b Sketch the graph of C versus t for $0 \leqslant t \leqslant 10$.
c What is the total cost when a bike is rented for 4 hours?
d If the total cost was \$50, for how many hours was a bike rented?

SOLUTION

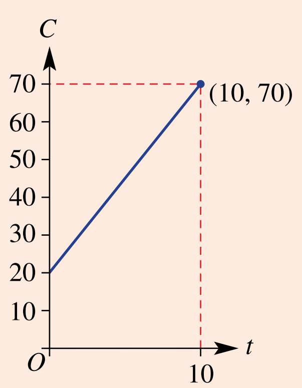
a $C = 20 + 5t$

A fixed amount of \$20 plus \$5 for each hour.

b C-intercept is at (0, 20)

At $t = 10$, $C = 20 + 5(10)$
$= 70$
$\therefore$ endpoint is (10, 70)

Let $t = 0$ to find the C-intercept.
Letting $t = 10$ gives $C = 70$ and this gives the other endpoint.



Sketch the graph using the points (0, 20) and (10, 70).

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

c $C = 20 + 5t$
$= 20 + 5(4)$
$= 40$
The cost is \$40.

Substitute $t = 4$ into the rule.

Answer the question using the correct units.

d $C = 20 + 5t$
$50 = 20 + 5t$
$30 = 5t$
$t = 6$
The bike was rented for 6 hours.

Write the rule and substitute $C = 50$. Solve the resulting equation for t by subtracting 20 from both sides then dividing both sides by 5.

Answer the question in words.

Now you try

A salesperson earns \$500 per week plus \$10 for each item sold.

a Write a rule for the total weekly wage, \$$W$, if the salesperson makes x sales.
b Draw a graph of W versus x for $0 \leqslant x \leqslant 100$.
c How much does the salesperson earn in a week if they make 25 sales?
d In a particular week the salesperson earns \$850. How many sales did they make?

Exercise 4J

FLUENCY	1–3	1, 2, 4	2, 4

Example 19

1 A sales representative earns \$400 a week plus \$20 for each sale she makes.
a Write a rule which gives the total weekly wage, \$$W$, if she makes x sales.
b Draw a graph of W versus x for $0 \leqslant x \leqslant 40$.
c How much does the sales representative earn in a particular week if she makes 12 sales?
d In a particular week, the sales representative earns \$1000. How many sales did she make?

2 A catering company charges \$500 for the hire of a marquee, plus \$25 per guest.
a Write a rule for the cost, \$$C$, of hiring a marquee and catering for n guests.
b Draw a graph of C versus n for $0 \leqslant n \leqslant 100$.
c How much would a party catering for 40 guests cost?
d If a party cost \$2250, how many guests were catered for?

3 A plumber charges a \$40 fee up-front and \$50 for each hour she works.
a Find a linear equation for the total charge, \$$C$, for n hours of work.
b What is the cost if she works 4 hours?
c If the plumber works on a job for two days and averages 6 hours per day, what will the total cost be?

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4 The cost, $\$C$, of recording a music CD is \$300, plus \$120 per hour of studio time.
  a Write a rule for the cost, $\$C$, of recording a CD requiring t hours of studio time.
  b Draw a graph of C versus t for $0 \leqslant t \leqslant 10$.
  c How much does a recording cost if it requires 6 hours of studio time?
  d If a recording cost \$660 to make, for how long was the studio used?

PROBLEM–SOLVING	5, 6	5, 6	6, 7

5 A petrol tank holds 66 litres of fuel. It contains 12 litres of petrol initially and the petrol pump fills it at 3 litres every 10 seconds.
  a Write a linear equation for the amount of fuel (F litres) in the tank after t minutes.
  b How long will it take to fill the tank?
  c How long will it take to add 45 litres into the petrol tank?

6 A tank is initially full with 4000 litres of water and water is being used at a rate of 20 litres per minute.
  a Write a rule for the volume, V litres, of water after t minutes.
  b Calculate the volume after 1.5 hours.
  c How long will it take for the tank to be emptied?
  d How long will it take for the tank to have only 500 litres?

7 A spa pool contains 1500 litres of water. It is draining at the rate of 50 litres per minute.
  a Draw a graph of the volume of water, V litres, remaining after t minutes.
  b Write a rule for the volume of water at time t minutes.
  c What does the gradient represent?
  d What is the volume of water remaining after 5 minutes?
  e After how many minutes is the pool half empty?

REASONING	8	8	8, 9

8 The rule for distance travelled d km over a given time t hours for a moving vehicle is given by $d = 50 + 80t$.
  a What is the speed of the vehicle?
  b If the speed was actually 70 km per hour, how would this change the rule? Give the new rule.

9 The altitude, h metres, of a helicopter t seconds after it begins its descent is given by $h = 350 - 20t$.
  a At what rate is the helicopter's altitude decreasing?
  b At what rate is the helicopter's altitude increasing?
  c What is the helicopter's initial altitude?
  d How long will it take for the helicopter to reach the ground?
  e If instead the rule was $h = 350 + 20t$, describe what the helicopter would be doing.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

ENRICHMENT: Sausages and cars	–	–	10, 11

10 Joanne organised a sausage sizzle to raise money for her science club. The hire of the barbecue cost Joanne \$20, and the sausages cost 40c each.

a **i** Write a rule for the total cost, $\$C$, if Joanne buys and cooks n sausages.

ii If the total cost was \$84, how many sausages did Joanne buy?

b **i** If Joanne sells each sausage for \$1.20, write a rule to find her profit, $\$P$, after buying and selling n sausages.

ii How many sausages must she sell to 'break even'?

iii If Joanne's profit was \$76, how many sausages did she buy and sell?

11 The directors of a car manufacturing company calculate that the set-up cost for a new component is \$6700 and each component would cost \$10 to make.

a Write a rule for the total cost, $\$C$, of producing x components.

b Find the cost of producing 200 components.

c How many components could be produced for \$13 000?

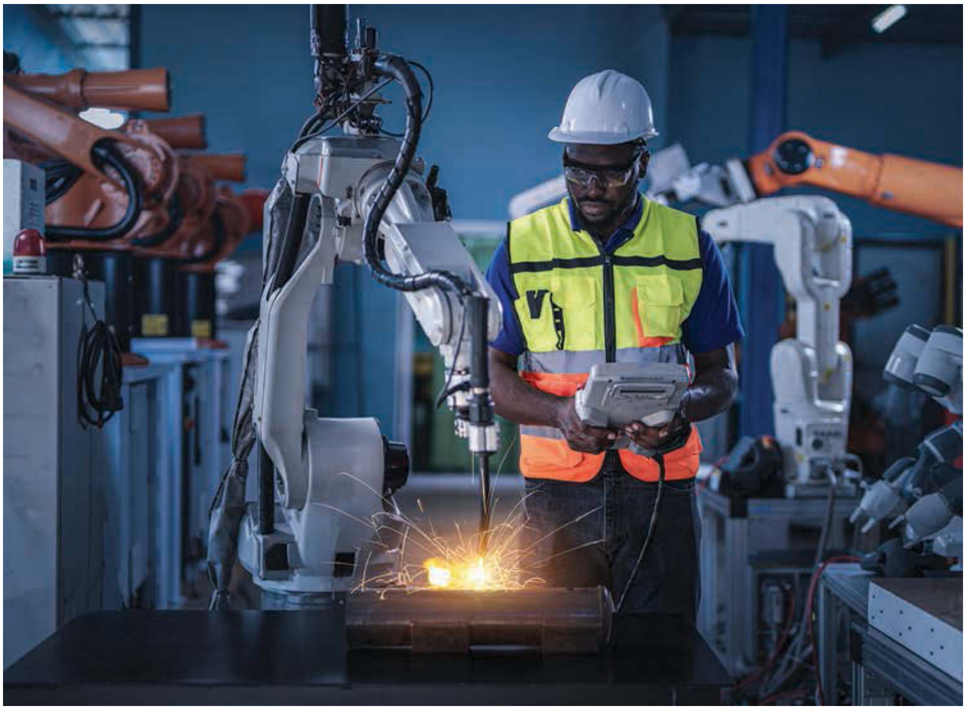
d Find the cost of producing 500 components.

e If each component is able to be sold for \$20, how many must they sell to 'break even'?

f Write a rule for the profit, $\$P$, in terms of x.

g Write a rule for the profit, $\$T$, per component in terms of x.

h Find x, the number of components, if the profit per component is to be \$5.



ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4K Graphical solutions to simultaneous equations

Learning intentions

- To understand that two straight line graphs intersect at exactly one point, unless the lines are parallel or identical
- To know that the point of intersection of two lines is the solution of the simultaneous equations
- To be able to use the graphs of two straight lines to read off the coordinates of the point of intersection
- To know how to use the equations to check if a point is at the intersection of the two lines

Past, present and future learning

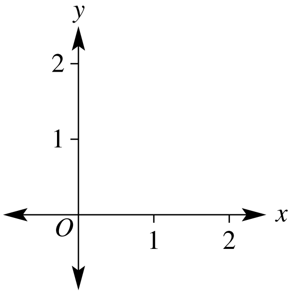
- This section addresses the Stage 5 Path Topic called Equations C
- These concepts will be revised, briefly, in Chapter 1 of our Year 10 book and also in Stage 6 Advanced

To find a point that satisfies more than one equation involves finding the solution to simultaneous equations. An algebraic approach was considered in Chapter 2. A graphical approach involves locating an intersection point.

Revenue, R, and costs, C, often have linear equations, such as $R = 1000n$ and $C = 400n + 3000$, for producing n surfboards. The break-even point of a business is where these two linear graphs intersect; profit starts as revenue begins to exceed costs.

Lesson starter: Accuracy counts

Two graphs have the rules $y = x$ and $y = 2 - 4x$.
Accurately sketch the graphs of both rules on a large set of axes like the one shown.

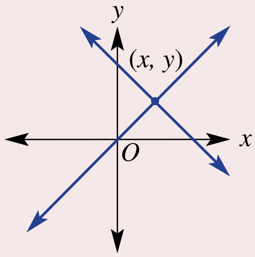


- State the x-value of the point at the intersection of your two graphs.
- State the y-value of the point at the intersection of your two graphs.
- Discuss how you could use the rules to check if your point is correct.
- If your point does not satisfy both rules, check the accuracy of your graphs and try again.

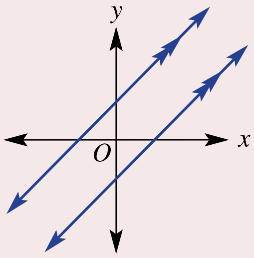
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

KEY IDEAS

- When we consider two or more equations at the same time, they are called **simultaneous equations**.
- To determine the **point of intersection** of two lines, we can use an accurate graph and determine its coordinates (x, y).
- Two situations can arise for the intersection of two different lines.
  - The two graphs intersect at one point only and there is one solution (x, y).
    - The point of intersection is simultaneously on both lines and is the **solution** to the simultaneous equations.



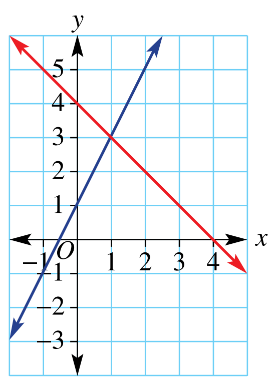
  - The two lines are parallel and there is no intersection.



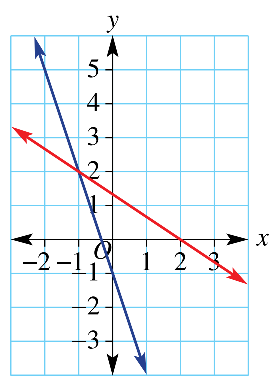
BUILDING UNDERSTANDING

1 Give the coordinates of the point of intersection for these pairs of lines.

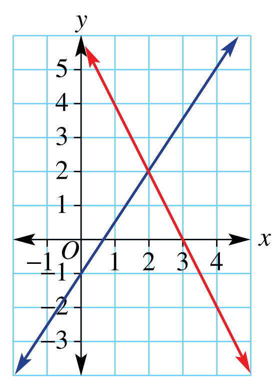
a



b



c



2 Decide if the point $(1, 3)$ satisfies these equations. (*Hint*: Substitute $(1, 3)$ into each equation to see if the equation is true.)

a $y = 2x + 1$ **b** $y = -x + 5$ **c** $y = -2x - 1$ **d** $y = 4x - 1$

3 Consider the two lines with the rules $y = 5x$ and $y = 3x + 2$ and the point $(1, 5)$.

a Substitute $(1, 5)$ into $y = 5x$. Does $(1, 5)$ sit on the line with equation $y = 5x$?

b Substitute $(1, 5)$ into $y = 3x + 2$. Does $(1, 5)$ sit on the line with equation $y = 3x + 2$?

c Is $(1, 5)$ the intersection point of the lines with the given equations?

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 20 Checking an intersection point

Decide if the given point is at the intersection of the two lines with the given equations.

a $y = 2x + 3$ and $y = -x$ with point $(-1, 1)$

b $y = -2x$ and $3x + 2y = 4$ with point $(2, -4)$

SOLUTION	EXPLANATION
a Substitute $x = -1$.	Substitute $x = -1$ into $y = 2x + 3$ to see if $y = 1$. If so, then $(-1, 1)$ is on the line.
$\begin{aligned} y &= 2x + 3 \\ &= 2 \times (-1) + 3 \\ &= -2 + 3 \\ &= 1 \end{aligned}$	
So $(-1, 1)$ satisfies $y = 2x + 3$.	
$\begin{aligned} y &= -x \\ &= -(-1) \\ &= 1 \end{aligned}$	Repeat for $y = -x$.
So $(-1, 1)$ satisfies $y = -x$. $\therefore (-1, 1)$ is the intersection point.	If $(-1, 1)$ is on both lines then it must be the intersection point.
b Substitute $x = 2$.	Substitute $x = 2$ into $y = -2x$ to see if $y = -4$. If so, then $(2, -4)$ is on the line.
$\begin{aligned} y &= -2x \\ &= -2 \times (2) \\ &= -4 \end{aligned}$	
So $(2, -4)$ satisfies $y = -2x$.	
$\begin{aligned} &3x + 2y \\ &= 3 \times (2) + 2 \times (-4) \\ &= 6 + (-8) \\ &= -2 \\ &\neq -4 \end{aligned}$	Substitute $x = 2$ and $y = -4$ to see if $3x + 2y = 4$ is true. Clearly, the equation is not satisfied.
So $(2, -4)$ is not on the line. $\therefore (2, -4)$ is not the intersection point.	As the point is not on both lines, it cannot be the intersection point.

Now you try

Decide if the given point is at the intersection of the two lines with the given equations.

a $y = 3x - 5$ and $y = -2x$ with point $(1, -2)$

b $y = -4x$ and $3x - y = 2$ with point $(2, -8)$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Example 21 Solving simultaneous equations graphically

Solve the simultaneous equations $y = 2x - 2$ and $x + y = 4$ graphically.

SOLUTION

$y = 2x - 2$

y-intercept (let $x = 0$):

$$\begin{aligned} y &= 2 \times (0) - 2 \\ y &= -2 \therefore (0, -2) \end{aligned}$$

x-intercept (let $y = 0$):

$$\begin{aligned} 0 &= 2x - 2 \\ 2 &= 2x \\ x &= 1 \therefore (1, 0) \end{aligned}$$

EXPLANATION

Sketch each linear graph by first finding the y-intercept (substitute $x = 0$) and the x-intercept (substitute $y = 0$) and solve the resulting equation.

$x + y = 4$

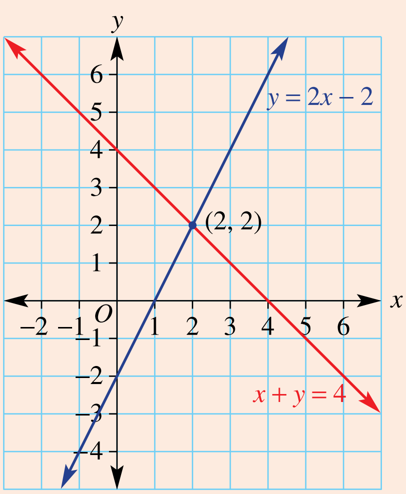
y-intercept (let $x = 0$):

$$\begin{aligned} (0) + y &= 4 \\ y &= 4 \therefore (0, 4) \end{aligned}$$

x-intercept (let $y = 0$):

$$\begin{aligned} x + (0) &= 4 \\ x &= 4 \therefore (4, 0) \end{aligned}$$

Repeat the process for the second equation.



Sketch both graphs on the same set of axes by marking the intercepts and joining with a straight line.

Ensure that the axes are scaled accurately.

Locate the intersection point and read off the coordinates.

The intersection point is (2, 2).
The solution is $x = 2,\ y = 2$.

The point (2, 2) simultaneously belongs to both lines.

Now you try

Solve the simultaneous equations $y = -x + 1$ and $2x + y = 3$ graphically.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Exercise 4K

FLUENCY	1, 2(½)	1–2(½)	1(½), 2(⅓)

Example 20

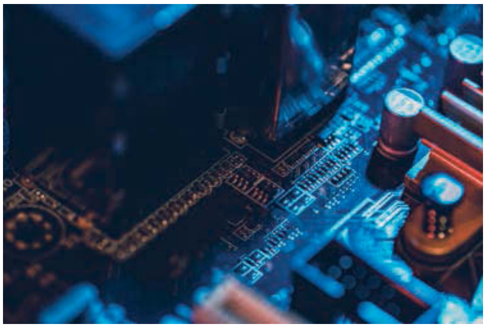
1 Decide if the given point is at the intersection of the two lines with the given equations.
- a $y = 3x - 4$ and $y = 2x - 2$ with point $(2, 2)$
- b $y = -x + 3$ and $y = -x$ with point $(2, 1)$
- c $y = -4x + 1$ and $y = -x - 1$ with point $(1, -3)$
- d $x - y = 10$ and $2x + y = 8$ with point $(6, -4)$
- e $2x + y = 0$ and $y = 3x + 4$ with point $(-1, 2)$
- f $x - 3y = 13$ and $y = -x - 1$ with point $(4, -3)$

Example 21

2 Solve these pairs of simultaneous equations graphically by finding the coordinates of the intersection point.
- a $2x + y = 6$, $x + y = 4$
- b $3x - y = 7$, $y = 2x - 4$
- c $y = x - 6$, $y = -2x$
- d $y = 2x - 4$, $x + y = 5$
- e $2x - y = 3$, $3x + y = 7$
- f $y = 2x + 1$, $y = 3x - 2$
- g $x + y = 3$, $3x + 2y = 7$
- h $y = x + 2$, $y = 3x - 2$
- i $y = x - 3$, $y = 2x - 7$
- j $y = 3$, $x + y = 2$
- k $y = 2x - 3$, $x = -1$
- l $y = 4x - 1$, $y = 3$

PROBLEM–SOLVING	3, 4	3, 4	4, 5

3 A company manufactures electrical components. The cost, $\$C$ (including rent, materials and labour), is given by the rule $C = n + 3000$, and its revenue, $\$R$, is given by the rule $R = 5n$, where n is the number of components produced.
- a Sketch the graphs of C and R on the same set of axes and determine the point of intersection.
- b State the number of components, n, for which the costs $\$C$ are equal to the revenue $\$R$.



4 BasketCo. manufactures baskets. Its costs, $\$C$, are given by the rule $C = 4n + 2400$, and its revenue, $\$R$, is given by the rule $R = 6n$, where n is the number of baskets produced. Sketch the graphs of C and R on the same set of axes and determine the number of baskets produced if the costs equal the revenue.

5 Two asteroids are 1000 km apart and are heading straight for each other. One asteroid is travelling at 59 km per second and the other at 41 km per second. How long will it be before they collide?

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

REASONING	6	6, 7	6–8

6 Explain why the graphs of the rules $y = 3x - 7$ and $y = 3x + 4$ have no intersection point.

7 For the following families of graphs, determine their points of intersection (if any).

a $y = x$, $y = 2x$, $y = 3x$

b $y = x$, $y = -2x$, $y = 3x$

c $y = x + 1$, $y = x + 2$, $y = x + 3$

d $y = -x + 1$ $y = -x + 2$, $y = -x + 3$

e $y = 2x + 1$, $y = 3x + 1$, $y = 4x + 1$

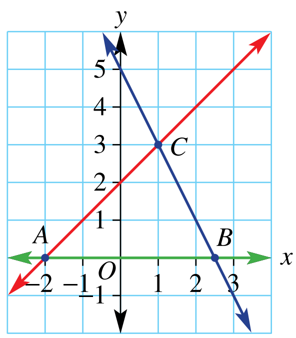
f $y = 2x + 3$, $y = 3x + 3$, $y = 4x + 3$

8 a If two lines have the equations $y = 3x + 1$ and $y = 2x + c$, find the value of c if the intersection point is at $x = 1$.

b If two lines have the equations $y = mx - 4$ and $y = -2x - 3$, find the value of m if the intersection point is at $x = -1$.

ENRICHMENT: Intersecting to find triangular areas	–	–	9, 10

9 The three lines with equations $y = 0$, $y = x + 2$ and $y = -2x + 5$ are illustrated here.



a State the coordinates of the intersection point of $y = x + 2$ and $y = -2x + 5$.

b Use $A = \frac{1}{2}bh$ to find the area of the enclosed triangle ABC.

10 Use the method outlined in Question 9 to find the area enclosed by these sets of three lines.

a $y = 0$, $y = x + 3$ and $y = -2x + 9$

b $y = 0$, $y = \frac{1}{2}x + 1$ and $y = -x + 10$

c $y = 2$, $x - y = 5$ and $x + y = 1$

d $y = -5$, $2x + y = 3$ and $y = x$

e $x = -3$, $y = -3x$ and $x - 2y = -7$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Working mathematically

Swimming at number 4

At a swim meet, Jess is the fourth and fastest swimmer in her team of 4 which swims the medley relay. Her average speed is 1.5 metres per second. During the race, she often finds herself in second position just before she dives in and needs to catch up so that she wins the race for her team. Each swimmer needs to swim 50 metres.

In the relay Adrianna is the last swimmer of another team which is currently in the lead before she dives in. To win Jess needs to catch up and overtake Adrianna. Adrianna's average speed is 1.4 metres per second.

For this task, d metres is the distance travelled by a swimmer and t seconds is the time in seconds after Adrianna dives into the water.

Present a report for the following tasks and ensure that you show clear mathematical workings and explanations where appropriate.

Routine problems

a Write a rule for Adrianna's distance (d metres) in terms of time (t seconds).

b Find how long it takes for Adrianna to complete the 50 metres to the nearest tenth of a second.

c If Jess dives into the water 2 seconds after Adrianna, then the rule describing Jess's distance after t seconds is given by: $d = 1.5t - 3$.

- **i** Find how long it takes for Jess to complete the 50 metres, after Adrianna dives into the water, to the nearest tenth of a second.
- **ii** Sketch a graph of d vs t for both Jess and Adrianna on the same set of axes. Show the coordinates of any endpoints and intercepts.
- **iii** Does Jess overtake Adrianna? If so, after how many seconds?

Non-routine problems

Explore and connect

a The problem is to determine the maximum head start that Adrianna can have so that Jess wins the relay for her team. Write down all the relevant information that will help solve this problem.

Choose and apply techniques

b Jess dives into the water 1 second after Adrianna. i.e. when $t = 1$ then $d = 0$ for Jess.

- **i** Find a rule for Jess's distance (d metres) after t seconds.
- **ii** Find the time taken for Jess to complete the 50 metres, after Adrianna dives into the water, to the nearest tenth of a second.
- **iii** Sketch a graph of d vs t for both Jess and Adrianna on the same set of axes. Show the coordinates of any endpoints and intercepts.
- **iv** Decide if Jess overtakes Adrianna. If so, after how many seconds?

c Repeat part **b** if Jess dives into the water 3 seconds after Adrianna.

Communicate thinking and reasoning

d If Jess catches up to Adrianna at exactly the time when Adrianna finishes the relay:

- **i** illustrate this situation with a graph
- **ii** determine the head start, in seconds to one decimal place, that Adrianna had in this situation.

e Decide under what circumstances will Jess win the relay for her team. Justify your answer.

f Summarise your results and describe any key findings.

Extension problems

Problem solve

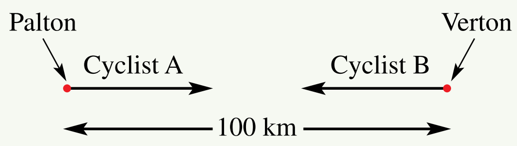
a If Jess's swim speed was able to be increased to 1.6 m/s, investigate the maximum head start that Adrianna could have so that Jess wins the relay.

b If in one particular race Adrianna has a 2.1 second head start, determine Jess's minimum speed if she is to win the relay.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Coming and going

The distance between two towns, Palton and Verton, is 100 km. Two cyclists travel in opposite directions between the towns, starting their journeys at the same time. Cyclist A travels from Palton to Verton at a speed of 20 km/h while cyclist B travels from Verton to Palton at a speed of 25 km/h.



Investigation

Measuring the distance from Palton

a Using d_A km as the distance cyclist A is from Palton after t hours, explain why the rule connecting d_A and t is $d_A = 20t$.

b Using d_B km as the distance cyclist B is from Palton after t hours, explain why the rule connecting d_B and t is $d_B = 100 - 25t$.

Technology: Spreadsheet

c Instructions:

- Enter the time in hours into column A, starting at 0 hours.
- Enter the formulas for the distances d_A and d_B into columns B and C.
- Use the **Fill down** function to fill in the columns. Fill down until the distances show that both cyclists have completed their journey.

	A	B	C
1	0	= 20 * A1	= 100 – 25 * A1
2	= A1 + 1		
3			

d **i** Determine how long it takes for cyclist A to reach Verton.
ii Determine how long it takes for cyclist B to reach Palton.
iii After which hour are the cyclists the closest?

Investigating the intersection

a Change the time increment to a smaller unit. Try 0.5 hours using '= A1 + 0.5', or 0.1 hours using '= A1 + 0.1' in column A.

b Fill or scroll down to ensure that the distances show that both cyclists have completed their journey.

c Determine the time at which the cyclists are the closest.

d Continue altering the time increment until you are satisfied that you have found the time of intersection of the cyclists correct to one decimal place.

e *Extension*: Complete part **d** above but find an answer correct to three decimal places.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Investigation

The graph

a Sketch a graph of d_A and d_B on the same set of axes. Scale your axes carefully to ensure that the full journey for both cyclists is represented.

b Determine the intersection point as accurately as possible on your graph and hence estimate the time when the cyclists meet.

c Use technology to confirm the point of intersection, and hence determine the time at which the cyclists meet, correct to three decimal places.

Algebra and proof

a At the point of intersection it could be said that $d_A = d_B$.
This means that $20t = 100 - 25t$.
Solve this equation for t.

b Find the exact distance from Palton at the point where the cyclists meet.

Reflection

Write a paragraph describing the journey of the two cyclists. Comment on:

- the speeds of the cyclists
- their meeting point
- the difference in computer and algebraic approaches in finding the time of the intersection point.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

1 A tank with 520 L of water begins to leak at a rate of 2 L per day. At the same time, a second tank is being filled at a rate of 1 L per hour starting at 0 L. How long does it take for the tanks to have the same volume?

2 Two cars travel towards each other on a 400 km stretch of road. One car travels at 90 km/h and the other at 70 km/h. How long does it take before they pass each other?

3 The points $(-1, 4)$, $(4, 6)$, $(2, 7)$ and $(-3, 5)$ are the vertices of a parallelogram. Find the midpoints of its diagonals. What do you notice?

4 A trapezium is enclosed by the straight lines $y = 0$, $y = 3$, $y = 7 - x$, and $x = k$, where k is a constant. Find the possible values of k given the trapezium has an area of 24 units^2.

5 Prove that the triangle with vertices at the points $A(-1, 3)$, $B(0, -1)$ and $C(3, 2)$ is isosceles.

6 Find the perimeter (to the nearest whole number) and area of the triangle enclosed by the lines with equations $x = -4$, $y = x$ and $y = -2x - 3$.

7 $ABCD$ is a parallelogram. A, B and C have coordinates $(5, 8)$, $(2, 5)$ and $(3, 4)$ respectively. Find the coordinates of D.

8 A kite is formed by joining the points $A(a, b)$, $B(-1, 3)$, $C(x, y)$ and $D(3, -5)$.

a Determine the equations of the diagonals BD and AC of this kite.

b Given $a = -5$ and $y = 4$, find the values of b and x.

c Find the area of the kite (without the use of a calculator).

9 A trapezium is enclosed by the straight lines $y = 0$, $y = 6$, $y = 8 - 2x$, and $y = x + k$, where k is a constant. Find the possible values of k given the trapezium has an area of 66 units^2.

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Chapter summary

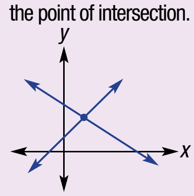
Linear relationships

Direct proportion

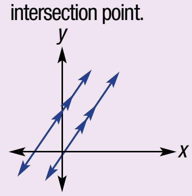
If two variables are directly proportional, their rule is of the form $y = mx$. The gradient of the graph represents the rate of change of y with respect to x; e.g. for a car travelling at 60 km/h for t hours, distance $= 60t$.

Graphical solutions of simultaneous equations

Graph each line and read off the point of intersection.



Parallel lines have no intersection point.

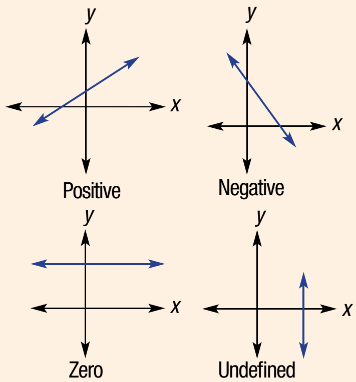


Linear modelling

Define variables to represent the problem and write a rule relating the two variables. The rate of change of one variable with respect to the other is the gradient.

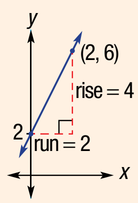
Gradient

This measures the slope of a line.



Gradient $m = \frac{\text{rise}}{\text{run}}$

$m = \frac{4}{2} = 2$



Finding the equation of a line

Gradient–intercept form ($y = mx + c$)

Require gradient and y-intercept or any other point to substitute; e.g. line with gradient 3 passes through point (2, 1)

$\therefore\ y = 3x + c$

substitute $x = 2$, $y = 1$

$1 = 3(2) + c$

$1 = 6 + c$

$\therefore\ c = -5$

$y = 3x - 5$

Straight line

$y = mx + c$

gradient y-intercept

In the form $ax + by = d$, rearrange to $y = mx + c$ form to read off gradient and y-intercept.

Midpoint and length of a line segment

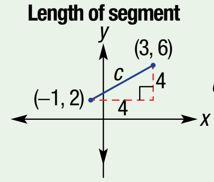
Midpoint is halfway between segment endpoints, e.g. midpoint of segment joining (–1, 2) and (3, 6).

$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$x = \frac{-1 + 3}{2} = 1$

$y = \frac{2 + 6}{2} = 4$, i.e. (1, 4)

Length of segment



$c^2 = 4^2 + 4^2$

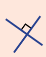
$c = \sqrt{32}$

$c = 5.7$ (to 1 d.p.)

Parallel and perpendicular lines

Parallel lines have the same gradient e.g. $y = 2x + 3$, $y = 2x - 1$ and $y = 2x$

Perpendicular lines are at right angles. Their gradients m_1 and m_2 are such that $m_1 m_2 = -1$, i.e. $m_2 = -\frac{1}{m_1}$.



A line perpendicular to $y = 2x + 3$ has gradient $-\frac{1}{2}$.

Linear relationship

A linear relationship is made up of points (x, y) that form a straight line when plotted.

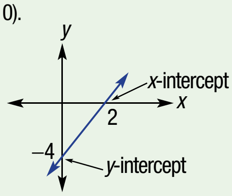
Sketching using intercepts

Sketch with two points. Often we find the x-intercept ($y = 0$) and y-intercept ($x = 0$).

e.g. $y = 2x - 4$

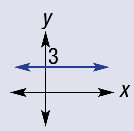
y-int: $x = 0$, $y = 2(0) - 4$
$= -4$

x-int: $y = 0$, $0 = 2x - 4$
$2x = 4$
$x = 2$

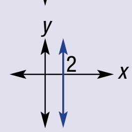


Special lines

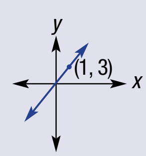
Horizontal line $y = c$
e.g. $y = 3$



Vertical line $x = k$
e.g. $x = 2$



$y = mx$ passes through the origin, substitute $x = 1$ to find another point, e.g. $y = 3x$



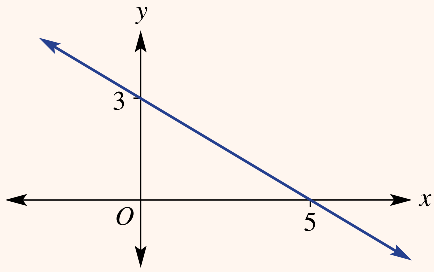
ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Chapter checklist with success criteria

A printable version of this checklist is available in the Interactive Textbook ✔

4A **1. I can plot points to graph a straight line.**
e.g. Using $-3 \leqslant x \leqslant 3$, construct a table of values and plot a graph of $y = 2x - 4$. ☐

4A **2. I can identify the *x*- and *y*-intercept.**
e.g. Write down the coordinates of the x- and y-intercepts from this graph.



☐

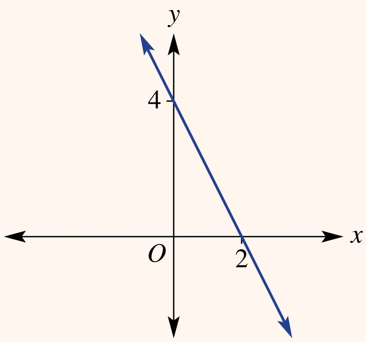
4A **3. I can determine if a point is on a line.**
e.g. Decide if the point $(-2, 9)$ is on the line with rule $y = -2x + 3$. ☐

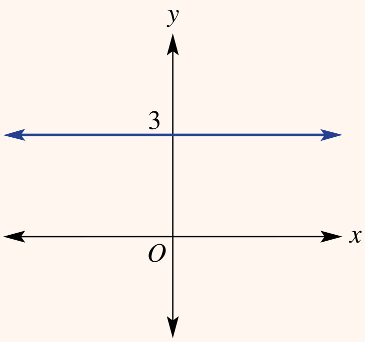
4B **4. I can sketch a graph of a straight line labelling intercepts.**
e.g. Sketch the graphs of $y = -2x + 4$ and $3x - 4y = 12$ showing x- and y-intercepts. ☐

4C **5. I can graph horizontal and vertical lines.**
e.g. Sketch $y = -2$ and $x = 5$. ☐

4C **6. I can sketch lines of the form $y = mx$.**
e.g. Sketch the graph of $y = 2x$. ☐

4D **7. I can find the gradient of a line.**
e.g. For the graphs shown, state whether the gradient is positive, negative, zero or undefined and find the gradient.





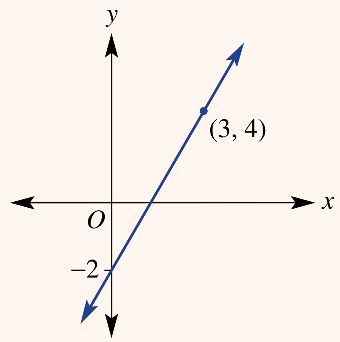
☐

4D **8. I can find the gradient between two points.**
e.g. Find the gradient of the line joining the points $(-1, 3)$ and $(4, 9)$. ☐

4E **9. I can work with direct proportion.**
e.g. A bathtub is being filled at a constant rate. It takes 10 minutes to fill the bath with 60 litres. Draw a graph of volume (V litres) vs time (t minutes) for $0 \leqslant t \leqslant 10$. Find the gradient of the graph and a rule for V. Use the rule to find the time for the bathtub to fill to 45 litres. ☐

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Chapter checklist

4F	**10. I can rearrange linear equations.** e.g. Rearrange $2x + 3y = 8$ to the form $y = mx + c$.	☐
4F	**11 I can state the gradient and *y*-intercept from a linear rule.** e.g. State the gradient and y-intercept of $y = -2x + 3$ and $2y - 4x = 5$.	☐
4F	**12. I can use the gradient and *y*-intercept to sketch a graph.** e.g. Use the gradient and y-intercept of $y = 3x - 2$ and $2x + 3y = 8$ to sketch their graphs.	☐
4G	**13. I can determine the equation of a straight line graph given the *y*-intercept.** e.g. Determine the equation of the straight line shown here. 	☐
4G	**14. I can determine the equation of a line given the gradient and a point.** e.g. Find the equation of the line which has a gradient of 2 and passes through the point $(-2, 5)$.	☐
4H	**15. I can find the midpoint of a line segment.** e.g. Find the midpoint $M(x, y)$ of the line segment joining the points $(2, -1)$ and $(6, 4)$.	☐
4H	**16. I can find the length of a line segment.** e.g. Find the length of the segment joining $(-2, 1)$ and $(3, 5)$, correct to two decimal places.	☐
4I	**17. I can find the equation of a parallel line.** e.g. Find the equation of a line that is parallel to the line $y = -2x + 3$ and passes through $(0, 1)$.	☐
4I	**18. I can find the equation of a perpendicular line.** e.g. Find the equation of a line that is perpendicular to the line $y = 4x + 2$ and passes through $(0, -3)$.	☐
4J	**19. I can apply linear relations.** e.g. The hire of a jumping castle involves a set-up cost of \$80 and a per hour rate of \$40. Write a rule for the total cost, $\$C$, of hiring a jumping castle for h hours and sketch its graph for $0 \leqslant h \leqslant 8$. Determine the total cost for hiring for 5 hours and how many hours it was hired for if the cost was \$200.	☐
4K	**20. I can check if a point is the intersection point of two lines.** e.g. Determine if the point $(2, -1)$ is at the intersection of the lines with equations $y = -2x + 3$ and $2x - 3y = 7$.	☐
4K	**21. I can solve simultaneous equations graphically.** e.g. Solve the simultaneous equations $y = 2x - 4$ and $x + y = 5$ graphically.	☐

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

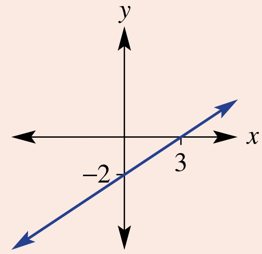
Short-answer questions

4A **1** Read off the coordinates of the x- and y-intercepts from the table and graph.

a

x	−2	−1	0	1	2
y	0	2	4	6	8

b



4B **2** Sketch the following linear graphs labelling x- and y-intercepts.

a $y = 2x - 4$ **b** $y = 3x + 9$ **c** $y = -2x + 5$ **d** $y = -x + 4$
e $2x + 4y = 8$ **f** $4x - 2y = 10$ **g** $2x - y = 7$ **h** $-3x + 6y = 12$

4B **3** Violet leaves her beach house by car and drives back to her home. Her distance d kilometres from her home after t hours is given by $d = 175 - 70t$.

a How far is her beach house from her home?
b How long does it take to reach her home?
c Sketch a graph of her journey between the beach house and her home.

4C **4** Sketch the following lines.

a $y = 3$ **b** $y = -2$ **c** $x = -4$
d $x = 5$ **e** $y = 3x$ **f** $y = -2x$

4D **5** By first plotting the given points, find the gradient of the line passing through the points.

a (3, 1) and (5, 5) **b** (2, 5) and (4, 3) **c** (1, 6) and (3, 1)
d (−1, 2) and (2, 6) **e** (−3, −2) and (1, 6) **f** (−2, 6) and (1, −4)

4E **6** An inflatable backyard swimming pool is being filled with water by a hose. It takes 4 hours to fill 8000 L.

a What is the rate at which water is poured into the pool?
b Draw a graph of volume (V litres) versus time (t hours) for $0 \leqslant t \leqslant 4$.
c Find the gradient of your graph and give the rule for V in terms of t.
d Use your rule to find the time to fill 5000 L.

4F **7** For each of the following linear relations, state the value of the gradient and the y-intercept, and then sketch using the gradient–intercept method.

a $y = 2x + 3$
b $y = -3x + 7$
c $2x + 3y = 9$
d $2y - 3x - 8 = 0$

4G **8** Give the equation of the straight line that:

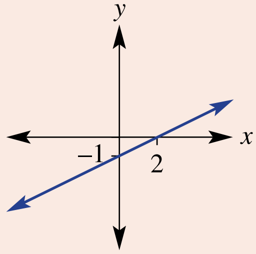
a has gradient 3 and passes through the point (0, 2)
b has gradient −2 and passes through the point (3, 0)
c has gradient $\frac{4}{3}$ and passes through the point (6, 3).

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

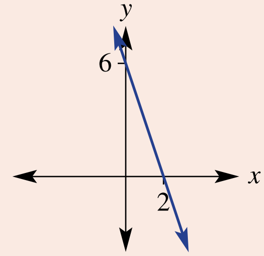
4G

9 Find the equations of the linear graphs below.

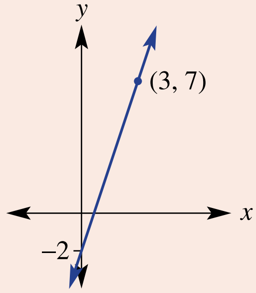
a



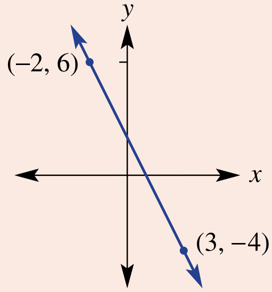
b



c



d



4H

10 For the line segment joining the following pairs of points, find:

i the midpoint

ii the length (to two decimal places where applicable).

a $(2, 4)$ and $(6, 8)$
b $(5, 2)$ and $(10, 7)$
c $(-2, 1)$ and $(2, 7)$
d $(-5, 7)$ and $(-1, -2)$

4H

11 Find the missing coordinate, n, if:

a the line joining $(-1, 3)$ and $(2, n)$ has gradient 2
b the line segment joining $(-2, 2)$ and $(4, n)$ has length 10, $n > 0$
c the midpoint of the line segment joining $(n, 1)$ and $(1, -3)$ is $(3.5, -1)$.

4I

12 Determine the linear equation that is:

a parallel to the line $y = 2x - 1$ and passes through the point $(0, 4)$
b parallel to the line $y = -x + 4$ and passes through the point $(0, -3)$
c perpendicular to the line $y = 2x + 3$ and passes through the point $(0, -1)$
d perpendicular to the line $y = -\frac{1}{3}x - 2$ and passes through the point $(0, 4)$
e parallel to the line $y = 3x - 2$ and passes through the point $(1, 4)$
f parallel to the line $3x + 2y = 10$ and passes through the point $(-2, 7)$.

4K

13 Determine if the point $(2, 5)$ is the intersection point of the graphs of the following pairs of equations.

a $y = 4x - 3$ and $y = -2x + 6$
b $3x - 2y = -4$ and $2x + y = 9$

4K

14 Find the point of intersection of the following straight lines using their graphs.

a $y = 2x - 4$ and $y = 6 - 3x$
b $2x + 3y = 8$ and $y = -x + 2$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

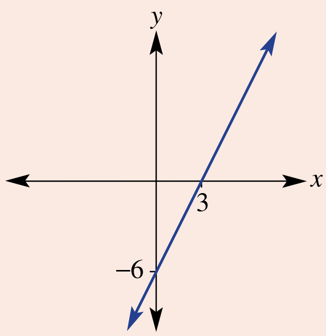
Multiple-choice questions

4A **1** The x- and y-intercepts of the graph of $2x + 4y = 12$ are respectively at:

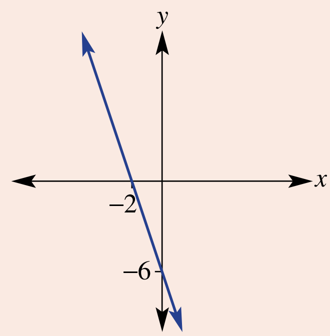
A (4, 0), (0, 2)
B (3, 0), (0, 2)
C (6, 0), (0, 3)
D (2, 0), (0, 12)
E (8, 0), (0, 6)

4B **2** The graph of $y = 3x - 6$ is represented by:

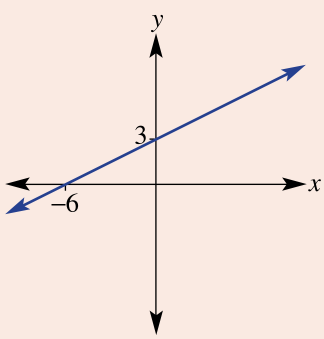
A



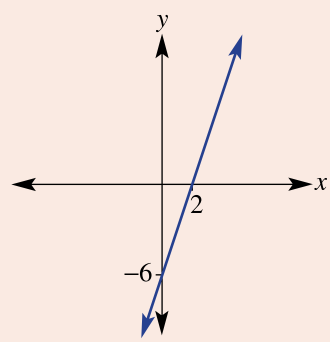
B



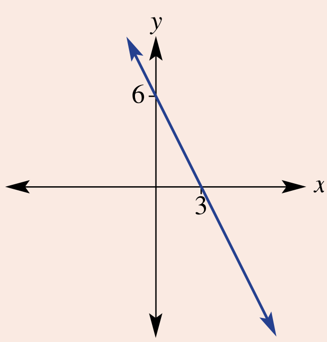
C



D



E



4D **3** The gradient of the line joining the points $(1, -2)$ and $(5, 6)$ is:

A 2
B 1
C 3
D -1
E $\frac{1}{3}$

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

Chapter review

4A

4 The point that is not on the straight line $y = -2x + 3$ is:

A $(1, 1)$
B $(0, 3)$
C $(2, 0)$
D $(-2, 7)$
E $(3, -3)$

4F

5 The linear graph that *does not* have a gradient of 3 is:

A $y = 3x + 7$
B $\frac{y}{3} = x + 2$
C $2y - 6x = 1$
D $y - 3x = 4$
E $3x + y = -1$

4G

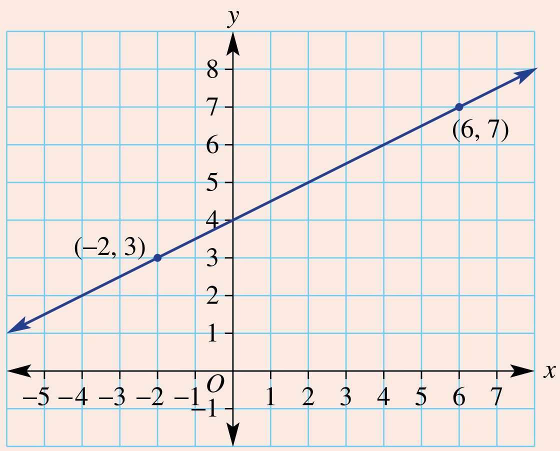
6 A straight line has a gradient of -2 and passes through the point $(0, 5)$. Its equation is:

A $2y = -2x + 10$
B $y = -2x + 5$
C $y = 5x - 2$
D $y - 2x = 5$
E $y = -2(x - 5)$

4G

7 The equation of the graph shown is:

A $y = x + 1$
B $y = \frac{1}{2}x + 4$
C $y = 2x + 1$
D $y = -2x - 1$
E $y = \frac{1}{2}x + 2$



4H

8 The length of the line segment joining the points $A(1, 2)$ and $B(4, 6)$ is:

A 5
B $\frac{4}{3}$
C 7
D $\sqrt{5}$
E $\sqrt{2}$

4I

9 The gradient of a line perpendicular to $y = 4x - 7$ would be:

A $\frac{1}{4}$
B 4
C -4
D $-\frac{1}{4}$
E -7

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.

4K

10 The point of intersection of $y = 2x$ and $y = 6 - x$ is:
- **A** (6, 2)
- **B** (−1, −2)
- **C** (6, 12)
- **D** (2, 4)
- **E** (3, 6)

Extended-response questions

1 Joe requires an electrician to come to his house to do some work. He is trying to choose between two electricians he has been recommended.
- **a** The first electrician's cost, $\$C$, is given by $C = 80 + 40n$ where n is the number of hours the job takes.
  - **i** State the hourly rate this electrician charges and his initial fee for coming to the house.
  - **ii** Sketch a graph of C versus n for $0 \leqslant n \leqslant 8$.
  - **iii** What is the cost of a job that takes 2.5 hours?
  - **iv** If the job costs \$280, how many hours did it take?
- **b** The second electrician charges a callout fee of \$65 to visit the house and then \$45 per hour.
  - **i** Give the equation for the cost, $\$C$, of a job that takes n hours.
  - **ii** Sketch the graph of part **b i** for $0 \leqslant n \leqslant 8$ on the same axes as the graph in part **a**.
- **c** Determine the point of intersection of the two graphs.
- **d** After how many hours does the first electrician become the cheaper option?

2 Abby has set up a small business making clay vases. Her production costs include a fixed weekly cost for equipment hire and a cost per vase for clay.
Abby has determined that the total cost of producing 7 vases in a week was \$146 and the total cost of producing 12 vases in a week was \$186.
- **a** Find a linear rule relating the production cost, $\$C$, to the number of vases produced, v. Start by plotting the given pair of points with the number of vases on the x-axis.
- **b** Use your rule to state:
  - **i** the initial cost of materials each week
  - **ii** the ongoing cost of production per vase.
- **c** At a selling price of \$12 per vase Abby determines her weekly profit to be given by $P = 4v - 90$.
  How many vases must she sell in order to make a profit?

ISBN 978-1-009-40936-0
© Palmer et al. 2024 Cambridge University Press
Photocopying is restricted under law and this material must not be transferred to another party.